

User Manual

for S32G PFE MCAL 4.4 Driver

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Chapter 1

Revision History

Revision	Date	Description
1.0	2021-04-21	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.0
1.1	2021-07-01	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.4
1.2	2021-10-12	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.5
1.3	2021-12-22	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.6
1.4	2022-02-04	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.7-CD01
1.5	2022-02-10	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.7-CD02
1.6	2022-04-07	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.7
1.7	2022-07-27	Prepared for release of S32G PFE MCAL 4.4 Driver Version 0.9.8
1.8	2022-10-12	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.0.0_CD1
1.9	2022-11-15	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.0.0
2.0	2023-05-15	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.1.0_CD2
2.1	2023-05-22	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.1.0
2.2	2023-07-27	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.2.0
2.3	2023-11-03	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.3.0_CD1
2.4	2023-12-15	Added documentation for errata, warnings and new features, compiler options documentation synchronized with RTD project, updated for driver version 1.3.0
2.5	2024-05-28	Prepared for release of S32G PFE MCAL 4.4 Driver Version 1.4.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductors' AUTOSAR Ethernet Driver for PFE on S32XX.

AUTOSAR Ethernet Driver configuration parameters description can be found in the Tresos Configuration Plugin section. Deviations from the specification are described in the [Deviations from Requirements](#) section.

AUTOSAR Ethernet driver requirements and APIs are described in the Ethernet Driver Software Specification Document (version 4.4.0).

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- S32G2XX, revision 2.0 and revision 2.1:
 - S32G274A_BGA525
 - S32G254A_BGA525
 - S32G233A_BGA525
 - S32G234M_BGA525
- S32G3XX, revision 1.1:
 - S32G378A_BGA525
 - S32G379A_BGA525
 - S32G398A_BGA525
 - S32G399A_BGA525
 - S32G338M_BGA525
 - S32G339M_BGA525
 - S32G358A_BGA525
 - S32G359A_BGA525

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
DEM	Diagnostic Event Manager
DET	Default Error Tracer
ETH	Ethernet
ETHIF	Ethernet Interface
ETHTRCV	Ethernet Transceiver
ETHSWT	Ethernet Switch
HIF	Host InterFace
ISR	Interrupt Service Routine
MCU	Micro Controller Unit
MII	Media Independent Interface
N/A	Not Available
PFE	Packet Forwarding Engine
PHY	PHYsical layer
RMII	Reduced Media Independent Interface
RAM	Random Access Memory
IDEX	Inter Driver EXchange

- The term "PFE" is related to the hardware module providing the Ethernet functionality.
- The term "Ethernet Controller" is related to a software interface providing standardized access to PFE.
- The term "Ethernet Driver" is related to the software handling the Ethernet Controllers and PFE.
- The term "Application" is used for the software utilizing the Ethernet Driver.
- The term "IDEX" is a request-response based communication protocol between PFE Driver instances.

2.5 Reference List

#	Title	Version
1	Specification of Ethernet Driver	AUTOSAR Release 4.↔ 4.0
2	Specification of Ethernet Interface	AUTOSAR Release 4.↔ 4.0
3	Requirements on Ethernet Support in AUTOSAR	AUTOSAR Release 4.↔ 4.0
4	S32G2 Reference Manual	Rev. 7, 02/2023
5	S32G3 Reference Manual	Rev. 4, 10/2023
6	FCI API Reference	Rev. 2.5.0
7	User Manual for S32 BASENXP Driver	Rev0000R4.0.2 Rev. 1.0
8	Integration Manual for S32 BASENXP Driver	Rev0000R4.0.2 Rev. 1.0
9	User Manual for S32 PLATFORM Driver	Rev0000R4.0.0 Rev. 1.0

#	Title	Version
10	Integration Manual for S32 PLATFORM Driver	Rev0000R4.0.0 Rev. 1.0
11	User Manual for S32 MCU Driver	Rev0000R4.0.0 Rev. 1.0
12	Integration Manual for S32 MCU Driver	Rev0000R4.0.0 Rev. 1.0
13	User Manual for S32 PORT Driver	Rev0000R4.0.0 Rev. 1.0
14	Integration Manual for S32 PORT Driver	Rev0000R4.0.0 Rev. 1.0
15	User Manual for S32 SERDES Driver	Rev0000R4.0.0 Rev. 1.0
16	Integration Manual for S32 SERDES Driver	Rev0000R4.0.0 Rev. 1.0
17	PFE Health Monitoring	Rev. 1.4.1, 12/2023

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.4.0 Specification of Ethernet Driver document [\[1\]](#)

3.2 Driver Design Summary

The Ethernet Driver controls the PFE engine of the S32G2X or S32G3X device. It provides the following features:

- Configuration and initialization of the PFE engine
- Loading, initialization and configuration of PFE firmware from external file
- Configuration and initialization of the Ethernet Controllers (the Ethernet Controllers are virtual as they are all mapped to single HW interface)
- Switching the Ethernet Controllers on and off
- Reception and transmission of Ethernet frames
- Access to the Ethernet Transceiver device registers through MDIO
- Access to the Ethernet Controller's RMON counters (RxStats, TxStats and TxErrorCounterValues)

- Access to PFE specific frame counters
- Handling of the PFE's interrupt requests
- Half and full duplex operation support
- AUTOSAR multicore type I (driver can be mapped to single core)
- Master-slave mechanism for running multiple drivers on one chip
- User mode support
- Switch driver management API for external switch
- Integration with Ethernet Transceiver driver (EthTrcv)
- Time Synchronization over Ethernet (The gPTP stack has to be provided by the upper layers)
- Time Aware shaper
- Checksum hardware offloading
- Internal L2 bridge and router among all PFE interfaces (EMACs and HIFs)
- FCI API for advanced runtime configuration
- Health Monitor for detection of faults and critical events at runtime

3.3 Hardware Resources

For the Ethernet driver, the main difference between S32G2X and S32G3X derivatives is in fault detection mechanism, which is covered in detail in [17]. There are no other differences in hardware resources between S32G2X and S32G3X derivatives.

The hardware module configured by the Ethernet driver is PFE. It has a single hardware instance.

The Ethernet controller to microcontroller pin mapping:

- For S32G2X derivatives it can be configured based on file **S32G2_IOMUX.xlsx**
- For S32G3X derivatives it can be configured based on file **S32G3_IOMUX.xlsx**

The files can be found attached to the Reference Manual.

3.4 Deviations from Requirements

3.4.1 Deviations from AUTOSAR requirements The driver deviates from the AUTOSAR Ethernet Driver Software Specification in some places.

The table [Status Column Description](#) identifies the AUTOSAR requirements that are not fully implemented, not implemented, or out of scope for the Ethernet Driver.

The table [Ethernet Requirements Deviations](#) provides the "Status" column description.

Term	Definition
N/S	Not In Scope
N/F	Not Fully Implemented
N/I	Not Implemented

Requirement	Status	Description	Notes
ECUC_Eth_00043	N/F	Configuration option EthCtrlConfig↔IngressFifoIdx shall be defined	Only one ingress fifo is supported per controller
ECUC_Eth_00042	N/F	Configuration option EthCtrlConfig↔IngressFifoPriorityAssignment shall be defined	Feature not implemented, option value ignored
SWS_Eth_00259	N/F	DRAFT: The module will operate as an independent instance in each of the partitions, means the called API will only target the partition it is called in	The requirement is implemented, but the maximum number of partitions is limited to one partition. That makes the PFE driver being multicore module type I, instead of type II, which is recommended for Ethernet driver by AUTOSAR.
SWS_Eth_00261	N/F	DRAFT: EthCtrlConfig, EthTrcv↔Config and EthSwtConfig (if existent in configuration) of one communication channel shall all reference the same ECUC partition	No checks were implemented in the EthCtrlConfig to enforce that.

3.4.2 Deviations from non-AUTOSAR requirements

3.4.2.1 Requirement AAVBR-781

A BSW module shall use the definitions provided by the BASE module for abstracting the: alignment/aligned declaration of variables, static(local) attributes, inline attributes, NULL_PTR

Deviation: Compiler keywords are being used directly without abstraction in file pfe_ct.h.

Rationale: File pfe_ct.h is being shared with PFE firmware (it defines API of the firmware), which is not an AUTOSAR component and for that reason it cannot use definitions from BASE module. It is ensured by testing that the compiler keywords are working correctly with all supported compilers and it is ensured by checks in code that no other compiler may be used for build.

3.4.2.2 Requirement AAVBR-537

All BSW modules shall not use compiler specific keywords.

Deviation: Files pfe_ct.h and pfe_compiler.h use compiler specific keywords directly.

Rationale: File pfe_ct.h uses compiler specific keywords because it is a file shared with PFE firmware (see rationale above). File pfe_compiler.h uses compiler specific keywords intentionally to check used compiler and its version, to avoid use of unsupported compiler.

3.5 Driver Limitations

List of known bugs, not implemented features and planned improvements is in the release notes.

Note

Driver was not tested running with AUTOSAR OS

3.5.1 Mitigation of Errata ERR051211 The PFE MCAL driver implements workaround for ERR051211. If option "Multiple PFE drivers support (master-slave)" is enabled and standard HIF is used, the workaround is enabled by default and it can be disabled by unselecting option "Enable errata ERR051211 workaround".

If the workaround is enabled, the driver will automatically allocate 72 extra Rx buffers and Rx buffer descriptors. The number 72 is a fixed value and there is no configuration option to change it.

If the workaround is enabled, configuration of TMU queue sizes for HIF interfaces (configurable through FCI command `FPP_CMD_QOS_QUEUE`) is limited by number of available Rx buffer descriptors. There are 2 queues available for each HIF and their default queue sizes are 16 and 16. When changing the values, their sum should not exceed 32. The FCI command will return error `FPP_ERR_QOS_QUEUE_SUM_OF_LENGTHS_EXCEEDED` if user attempts to set the values too high (with respect to available Rx buffers).

3.5.2 Mitigation of Errata ERR051699 Due to the errata ERR051699, PFE EMAC can reliably work in 10Mbps MII/RMII mode only with compatible PHYs (see the workaround below).

MII is nibble-based (4b) interface and RMII is dibble based (2b). When operating at 10 Mbps, PHYs may remove bits (multiples of 2b for RMII, 4b for MII), leaving the packet non-octet aligned. However, the EMAC supports only octet-aligned packets.

Workaround

1. Use 100Mbps operation instead of 10Mbps, and use PHY that guarantees octet-only alignment for 100Mbps MII/RMII,
2. or use a PHY that supports octet-only alignment for 10 Mbps.

3.5.3 Checksum Offload Limitation The checksum offload is limited to following combinations only:

- IPV4+TCP
- IPV4+UDP
- IPV4_OPTIONS+TCP
- IPV4_OPTIONS+UDP
- IPV6+TCP
- IPV6+UDP
- IPV6_EXT+TCP

- IPV6_EXT+UDP
- IPV4+TCP with single tag with TPID 8100
- IPV4+UDP with single tag with TPID 8100
- IPV6+TCP with single tag with TPID 8100
- IPV6+UDP with single tag with TPID 8100
- PPPoE+IPV4+TCP
- PPPoE+IPV4+UDP
- PPPoE+IPV4_OPTIONS+TCP
- PPPoE+IPV4_OPTIONS+UDP
- PPPoE+IPV6+TCP
- PPPoE+IPV6+UDP
- PPPoE+IPV6_EXT+TCP
- PPPoE+IPV6_EXT+UDP
- PPPoE+IPV4+TCP with single tag with TPID 8100
- PPPoE+IPV4+UDP with single tag with TPID 8100
- PPPoE+IPV6+TCP with single tag with TPID 8100
- PPPoE+IPV6+UDP with single tag with TPID 8100
- IPV4+IPV4+ICMP

Note

Checksum offload for double-VLAN-tagged frames and VLAN-tagged frames with TPID different from 8100 is not supported.

Checksum offload for fragmented frames is not supported.

3.5.4 Frame replicas share modifications Frame replicas use data of the original (parent) ingress frames. Any modification of an ingress frame is propagated to all replicas of the frame¹ (if there are any replicas); even to the replicas which were created **before** the said that modification occurred.

This limitation applies to all frame data: MAC addresses, VLAN tags (see below), IP addresses, ports, payload, etc.

VLAN tag modifications

PFE can *add*, *delete*, or *change* the outer VLAN tag of each individual frame/replica. In one particular scenario, these modifications trigger the limitation in a specific way. See the table below. In the table, the term ITEM represents both an original ingress frame and any replica of the given frame.

Scenario which triggers the limitation (all points must apply):

- The original ingress frame has 1 or more VLAN tags.
- Some ITEMS (but not all) are subjected to VLAN tag modification *change* or *delete* (ITEMS_A).
- The rest of the ITEMS is either subjected to VLAN tag modification *add*, or is left "as is", without any VLAN tag modification (ITEMS_B).

Effect of the limitation in the scenario:

After modifications, all ITEMS_B will be missing the VLAN tag of the original ingress frame.

Note

This limitation is most visible during RX mirroring:

1. Ingress frame arrives to PFE.
2. RX replica of the frame is allocated. For performance reasons, it is not sent immediately.
3. The frame is classified and optionally modified by PFE. Almost all modifications of the frame are also propagated to the already existing RX replica. If the frame is subjected to VLAN tag modification "change" or "delete", then the RX replica can even lose the VLAN tag of the original ingress frame.
4. The frame and all its replicas are egressed from PFE.

Remarks

[1] Similarly, any modification of a replica is propagated to all its sister replicas and to the original ingress frame.

3.5.5 QoS known limitations

1. Egress QoS Scheduler configuration is not fully dynamic

affected FCI commands: FPP_CMD_QOS_SCHEDULER + FPP_ACTION_UPDATE

Description: It is not safe to configure properties of Egress QoS Scheduler once any sort of traffic is processed by PFE. An attempt to configure properties of Egress QoS Scheduler once any traffic has passed through PFE can lead to temporary malfunction of Egress QoS Scheduler.

Workaround: Configure Egress QoS Schedulers immediately after driver initialisation and avoid doing traffic over EMACs before that.

2. Ingress QoS configuration not dynamic

affected FCI commands: FPP_CMD_QOS_POLICER + FPP_ACTION_UPDATE

Description: Traffic is sometimes blocked on reception at EMAC level when the Ingress QoS (IQOS) module is enabled, and resumes after disabling the IQOS module. The issue is triggered in the following cases: when IQOS is re-enabled multiple times (same initial PFE driver load), or when IQOS is enabled after the PFE driver had significant traffic over the corresponding EMAC. The issue is caused by the fact that 'dynamic' configuration of the Ingress QoS module is not reliable.

Workaround: Enable the Ingress QoS module once immediately after driver initialisation, and before having traffic over the corresponding EMAC.

3. Ingress QoS Shaper (policer) may block Rx traffic after burst of 133 to 140 byte frames (L2) on a 2.5G link

affected FCI commands: FPP_CMD_QOS_POLICER_SHP + FPP_ACTION_UPDATE

Description: The following issue can be triggered with a burst (i.e. 50) of basic L2 frames (i.e. both IPv4 and IPv6, no VLAN tags) injected close to line rate on the 2.5 Gbps link, after enabling the Ingress QoS Shaper for a 2.5G PFE port: the GPI hardware watchdog timer is triggered and reported by Health Monitor (HM) module, followed by Rx traffic being blocked for the given Rx port until full PFE reset. The issue can be triggered only by frames ranging from **133 bytes to 140 bytes** (inclusive) in L2 size, including the 4B FCS. The issue has been observed for ingress burst rates of 2.2 Gbps and above. So far the issue could not be triggered with lower ingress packet rates for 2.5G links, nor for 1G links (or lower).

Workaround: This is a hardware limitation with no available workaround to date.

4. Ingress QoS Shaper (policer) output rate is half the expected rate for some input streams

affected FCI commands: FPP_CMD_QOS_POLICER_SHP + FPP_ACTION_UPDATE

Description: The output rate of the PFE Ingress QoS hardware shaper (policer) is roughly half (50%) of the expected rate, provided as `isl` (idle slope) parameter to the shaper configuration command, for an input stream comprised of L2 frames that are bigger than **132 bytes**, including the 4B FCS. For frames that are up to (including) 132 B the shaper output rate is correct.

Workaround: This is a hardware limitation with no available workaround to date.

5. Ingress QoS does not support ECN (RFC 3168) classification via the 8 bit `tos` parameter

affected FCI commands: FPP_CMD_QOS_POLICER_FLOW + FPP_ACTION_REGISTER

Description: Despite of taking in an 8-bit value as the input parameter called `tos`, the Ingress QoS classifier does only classification based on the 6-bit DSCP field, as specified by RFC2474. The lower 2 bits of the `tos` argument are ignored by the Ingress QoS classifier.

Workaround: This is a hardware limitation. Use only the upper 6-bit DSCP sub-field for flow classification.

6. WRED "full" threshold not configurable

affected FCI commands: FPP_CMD_QOS_POLICER_WRED + FPP_ACTION_UPDATE

Description: The WRED "full" threshold (`.thr[FPP_IQOS_WRED_FULL_THR]`) is not configurable. The threshold at which all the ingress traffic is dropped is given by the actual queue size (i.e. DMEM or LMEM). Configuring lower values for `.thr[FPP_IQOS_WRED_FULL_THR]` than the ingress queues sizes has no effect.

Workaround: Ignore the `.thr[FPP_IQOS_WRED_FULL_THR]` parameter, and WRED will assume by default the maximum possible size of the associated ingress queue.

3.6 Driver usage and configuration tips

3.6.1 FCI API The driver provides FCI API, which can be enabled in Tresos by option "Enable FCI, routing and bridging support". All FCI API calls are handled by master driver. The FCI API is available also on slave driver, which connects to master driver through FCI proxy. The FCI API must not be used on master and slave driver at the same time.

Note

The FCI API is documented in separate document [6] FCI API Reference, which is delivered with this release.

3.6.2 AUX Controller The AUX interface is not attached to any EMAC and it can be used to send/receive Ethernet frames to/from any HIF or EMAC interface through bridge or other packet forwarding mechanism.

In order to configure the AUX interface in EB Tresos add one new Ethernet controller and for the option "EthCtrl←Port" select the "AUX" value.

In case of bridge configuration where an EMAC is part of a bridge domain and also configured in a controller, the frames received on the EMAC from outside the SoC and forwarded to the driver's HIF according to bridging/routing rules can be processed by the driver in 2 ways:

1. All the frames are received by the EMAC controller.
2. Only management frames are received on the EMAC controller, all normal frames are received on the AUX controller.

If parameter "Accept all traffic" is enabled (this is the default), process 1 will be used by the driver. If disabled, process 2 will be used. It is recommended to disable this parameter when the corresponding EMAC is part of a bridge domain. This allows the upper layers to send and receive all bridge-related traffic on the same (AUX) controller.

Option 'Accept all traffic' is available for each EMAC controller. The option has no effect if the EMAC is not used in bridging/routing.

AUX Ethernet controller shall have the MAC address configured, because in bridge operation modes the frames are forwarded among physical interfaces based on MAC addresses.

Bridging or routing can be configured on the HIF and EMAC interfaces with FCI API, see [6].

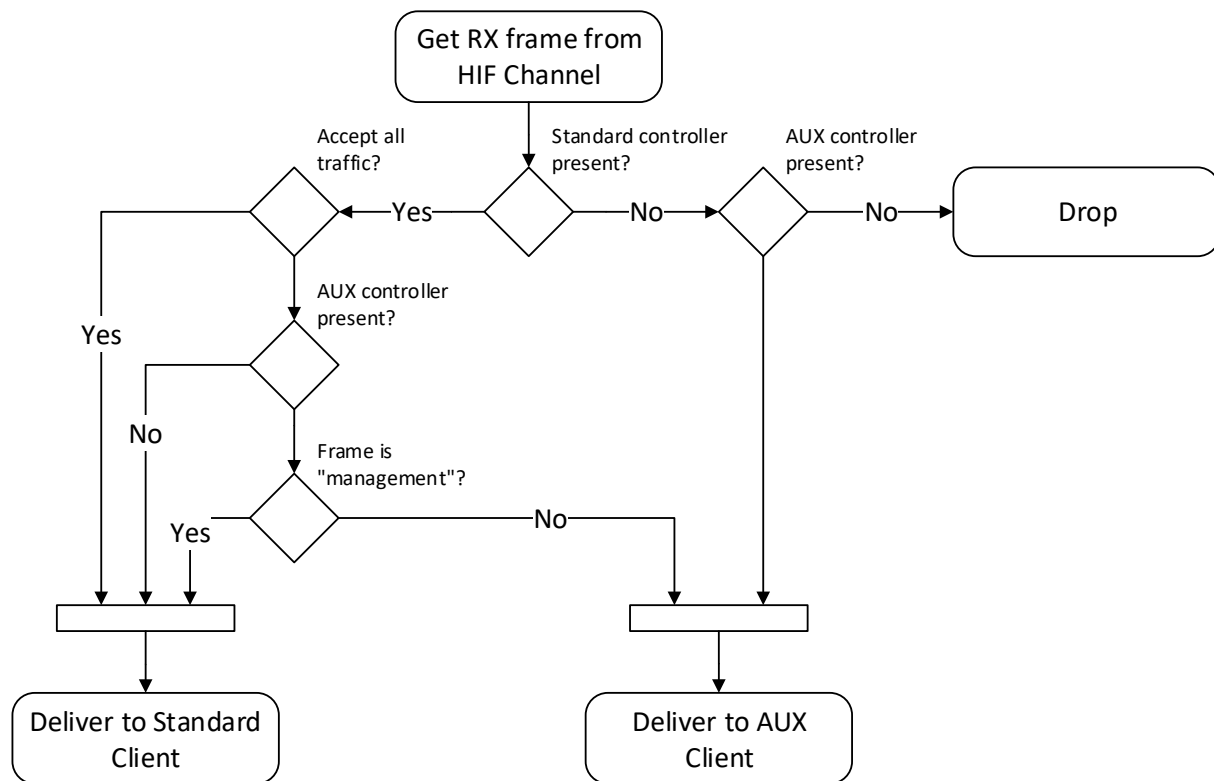


Figure 3.1 Ingress packet dispatching in HIF driver. Standard Client relates to EMAC Controller

3.6.3 HIF Controller The driver provides the capability to attach to a "remote" Host Interface (HIF). A remote HIF refers to a HIF that is not utilized by the current driver instance for host communication, as it is owned by another driver instance. This feature enables the sending and receiving of Ethernet frames to and from any other controller instances connected to a different HIF interface. It's important to note that such controllers are not attached to any EMAC, so all MAC related options are disabled for HIF controller.

This functionality establishes a communication datapath for the upper layers, facilitating data exchange across different platforms that share the same Packet Forwarding Engine (PFE) periphery.

To attach a controller to HIF0, HIF1, HIF2, HIF3, or HIF_NOCPY remote interfaces in EB Tresos, select the corresponding option in the "EthCtrlPort" parameter.

During transmission, the controller attached to the HIF operates in TX_INJECT mode. In this mode, the driver labels the custom HIF Tx header of each packet to instruct the PFE to bypass the classification stage and directly send the packets to the egress HIF port configured in the custom HIF Tx packet header.

The diagram below illustrates the traffic flow between two network interfaces bound to HIFs, representing separate driver instances:

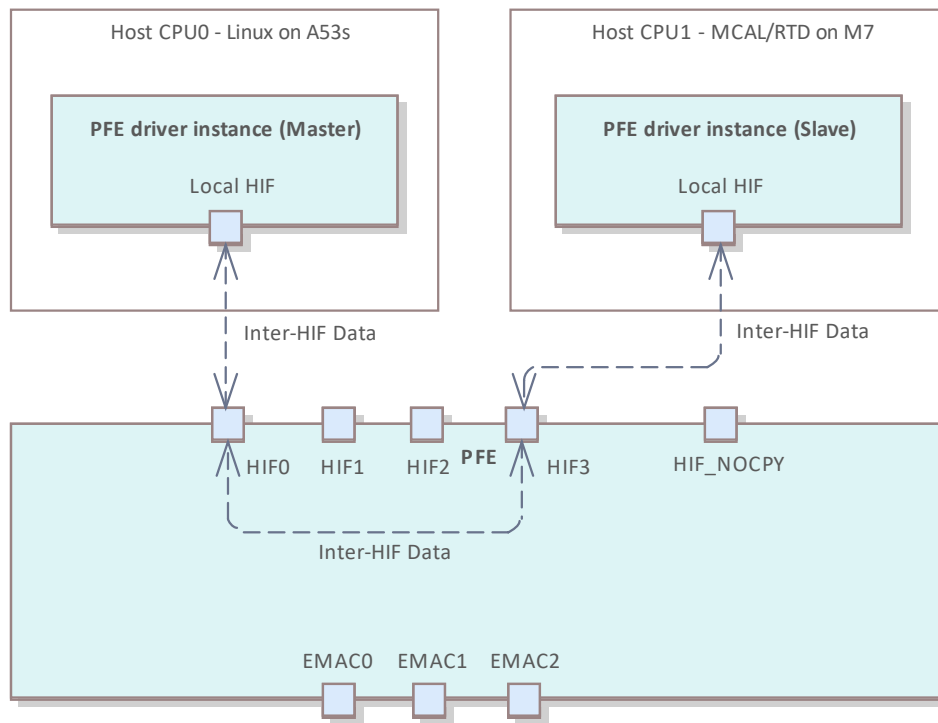


Figure 3.2 HIF to HIF communication

3.6.4 Master-Slave Runtime Facts The driver can run in multiple instances within the same system to allow sharing the PFE-provided connectivity using dedicated host interfaces. The intended use case is to run a master instance within one Host, and one or more slave instances from other Hosts that can have access to PFE, even from other operating systems than the Host OS running the master instance. This feature can be controlled by the "Slave driver" option resulting in build either of master or slave driver variant.

3.6.4.1 Driver Start-up Synchronization

A crucial aspect in running a master-slave scenario is to ensure that the driver instances synchronize correctly at start-up. The slave should not start its hardware initialization before the master is functional, otherwise the PFE can be rendered non-functional. To prevent this, a Host/OS independent synchronization mechanism is build within the PFE driver. The mechanism uses global system registers as mailboxes for signaling between the master and slave instances, and the synchronization is done in two steps: the 'IP ready' step, and the 'master initialization ready' step. Bit 16 of GENCTRL3 register (address 0x4007CAEC) is used by PFE master to signal 'IP ready' to slaves. The MCAL slave driver implements blocking wait for both steps in function `Eth_43_PFE_Init`. Wait for master can be controlled by option "EthSlaveHifMasterUpTimeout" in slave driver configuration.

3.6.4.2 Driver Reset

When master driver is reset, all slave drivers must be reset as well. For more details refer to [17].

3.6.4.3 Packet Distribution among Drivers

Starting with version 1.4.0 of the MCAL PFE Ethernet driver, automatic creation of logical interfaces (LOG_IFs) by the driver is limited to creation of default logical interfaces only. During initialization, the master driver creates one default logical interface for each physical interface (PHY_IF: EMAC, HIF or HIF_NOCPY). The names of the default LOG_IFs match the names of associated PHY_IFs (e.g. `emac0`, `hif3`, `hifncpy`). Slave drivers don't create any LOG_IFs. If needed, user can create additional LOG_IFs, or change configuration of default LOG_IFs via the FCI API (See [6]).

Note

Purpose of the default LOG_IF is for forwarding frames whose target is not specified by any other means, such as: TX_INJECT flag, bridging or a matching route. Purpose of additional LOG_IFs, if created by user, is to be used for flexible router configuration.

TX_INJECT flag is automatically applied to traffic originated from hosts, if the traffic is sent through an E↔MAC or a HIF controller. AUX controller does not set the TX_INJECT flag. Traffic received through EMACs also does not have the TX_INJECT flag set. The TX_INJECT flag overrides any routing or bridging.

The main benefit of this consolidation is that the PFE drivers no longer interfere with user configuration of forwarding mechanisms in PFE. Mainly:

- drivers don't change the operational modes of the PHY_IFs automatically. After the master driver initialization, all PHY_IFs are in default operational mode. The user has full control, using the FCI API to configure the operational modes if needed.
- drivers don't create LOG_IFs ad-hoc. After the master driver initialization, the user has full control over LOG_IFs, using the FCI API to configure them if needed.

The main drawback of the consolidation is that, after initialization, slave drivers cannot receive any traffic from EMACs, even if they have EMAC controllers configured. The reason is that all default LOG_IFs are by default configured to forward all traffic to the Master driver. All drivers can transmit to any PHY_IF using TX_INJECT flag (if they have respective controllers configured).

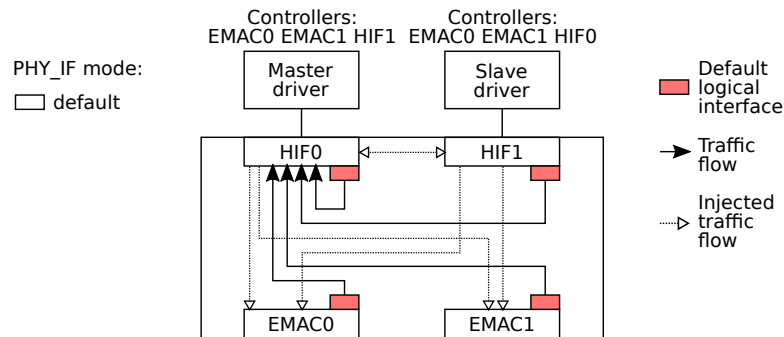


Figure 3.3 Default configuration of PFE. Slave drivers can receive only traffic with TX_INJECT flag sent from other HIFs

To mitigate this limitation, user can use the FCI API to configure preferred packet distribution mechanism. For example:

a) Configure some form of bridging or routing. This is the recommended approach for most applications.

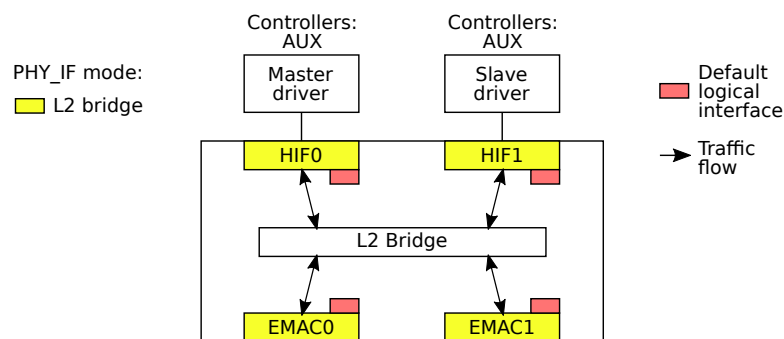


Figure 3.4 Example L2 bridge configuration

b) Change the target of default LOG_IF(s). This is not needed for master driver (it is the target already), but it is needed for a slave driver if it wants to receive frames from certain PHY_IF(s). Note that this change does not enable sharing of one EMAC among multiple drivers.

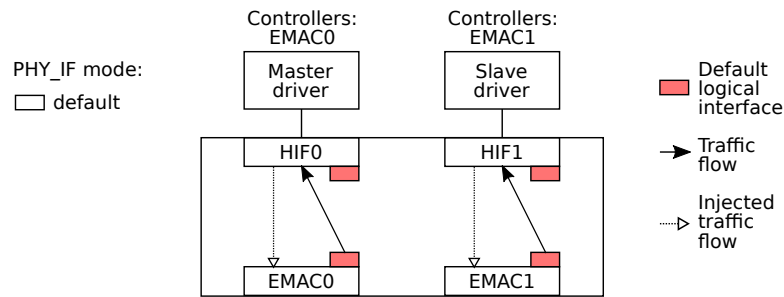


Figure 3.5 Example of 2 drivers. Each of them is using one EMAC.

c) Restore legacy behavior: configure flexible router mode IF_OP_FLEX_ROUTER on EMAC(s) and create additional LOG_IFs to configure forwarding of frames. Note that this would need to be done also for multicast addresses as function `Eth_43_PFE_UpdatePhysAddrFilter` does not create the LOG_IFs automatically (previously it did). This approach suffers from broadcast and multicast related limitations and for most use-cases it is not recommended.

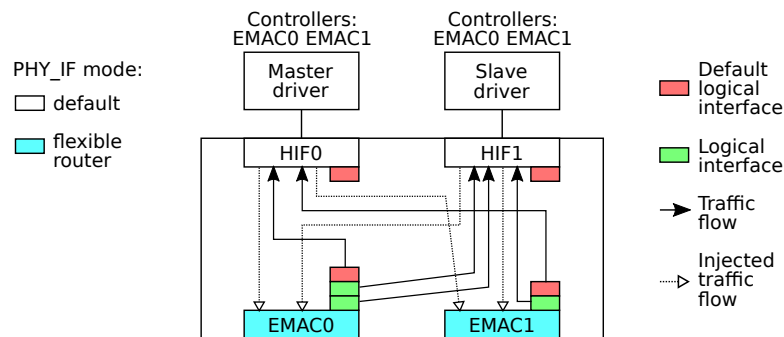


Figure 3.6 Example of 'legacy' configuration, where two drivers are sharing two EMACs. Logical interfaces forward selected traffic to the Slave driver. Default logical interfaces forward remaining traffic to the Master driver.

d) Combine various approaches.

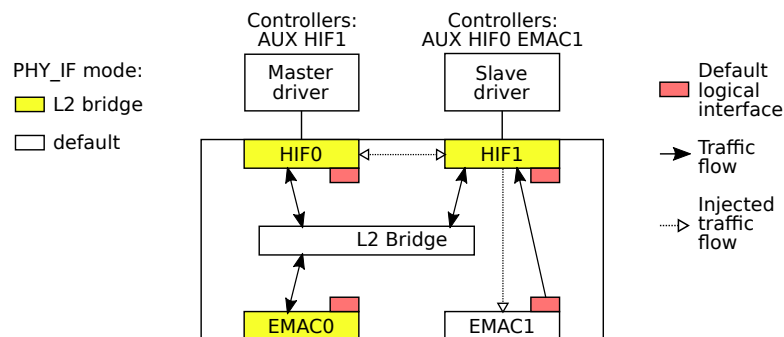


Figure 3.7 Example of two drivers connected to each other and to EMAC0 through a bridge, while the Slave driver also uses EMAC1, while the two drivers also have a direct HIF to HIF connection between them.

3.6.5 Health Monitor Faults and other critical events detected in PFE hardware, firmware, or driver are reported both through DEM module and through Health Monitor API (part of FCI API). The events reported through DEM and Health Monitor are the same (except standard AUTOSAR events, which are reported only to DEM) and they can be mapped based on their name. The main difference is that Health Monitor API provides more information about detected events and it is consistent across all PFE drivers (AUTOSAR, Linux, QNX).

Refer to PFE Health Monitoring manual [17], which describes in detail the Health Monitor API, events and recovery actions. The recovery actions are applicable even if DEM is used to detect the events.

3.6.6 Ungraceful HIF Channel Reset The PFE slave driver instance may face ungraceful termination scenarios, necessitating a robust recovery mechanism. During such shutdowns, the PFE hardware remains unaware of the abrupt termination and continues with data transfers. This can potentially result in unintended data leaks or corruption if not adequately addressed.

To mitigate this issue, during driver initialization:

1. The master driver will always restart the PFE HW. In this case all slave drivers need to be restarted as well.
2. The slave driver will check its HIF channel registers for being previously initialized and, if initialized, the driver will automatically execute the ungraceful HIF channel reset procedure. This allows the slave driver instance to be reset independently without having to reset other PFE drivers. The ungraceful HIF channel reset procedure will not be executed if memory allocation has changed from previous run.

The PFE master driver instance resets PFE hardware directly so the procedure is not applied there.

Procedure Overview: The ungraceful HIF channel reset procedure withing a slave driver instance comprises four main steps:

1. Find TX Index: This step involves locating the active transmission buffer descriptor, which is needed to enable resuming transmissions in the PFE driver
2. Find RX Index: Similarly, active reception buffer descriptor is found, enabling the driver to resume receiving data
3. Sync RX BD Ring to Head: The RX BD ring is synchronized to the head, ensuring expected state after driver initialization is done
4. Sync TX BD Ring to Head: The TX BD ring is also synchronized to the head, ensuring expected state after driver initialization is done

By following this ungraceful HIF channel reset procedure, the PFE slave driver can effectively recover from abrupt shutdowns and maintain the integrity of data transfers, ensuring a reliable and secure packet forwarding interface periphery.

Note

Due to a missing dependency for the Slave ungraceful reset procedure the Slave HIF ungraceful reset can fail. To prevent the failure, the integrator must perform an extra step in a situation when the Slave driver instance crash was detected. It must be ensured that corresponding Slave HIF port (PHY_IF) is disabled, the corrective action must be executed in the Master instance before the Slave module is reloaded.

3.6.7 Resource allocation The configuration parameter *VarEthBmu2BufCnt* (number of available memory buffers) needs to be configured sufficiently large so that the pool is not depleted by the configured Egress QoS queues even in a burst load scenario on any single interface.

There is a total of eight Egress QoS queues available for every configured interface with the default lengths being:

- queue 0: 150 entries
- queues 1-7: 13 entries

The queues' lengths can be re-configured at run-time via the FCI interface (*FPP_CMD_QOS_QUEUE*).

It is necessary that the configured value of *VarEthBmu2BufCnt* exceeds the size of the largest individual Egress QoS queue for any configured interface (i.e. greater than 150 in the default case). Otherwise, the buffer pool could be depleted even before the maximum queue length is reached.

Please note that, to guarantee, that a single interface never blocks the whole system, the *VarEthBmu2BufCnt* parameter would have to exceed the worst-case sum of lengths of all Egress QoS queues for any configured interface. This may not be always practical due to limited available memory and a trade-off needs to be found between system throughput and the number of allocated buffers.

3.6.8 IDEX IDEX is a request-response communication protocol between multiple PFE driver instances. The IDEX applies to multiclient driver mode only, where different driver instances exchange messages through HIF channels using IHC. The IDEX allows the Slave Driver to forward configuration requests that the Master Driver instance has an exclusive access to.

The following options can be used to configure IDEX communication on the Slave driver:

PfeIdxRpcResendTime This option specifies the maximum response wait time after sending a request. If no response is received within this time, a request is retransmitted or the function times out if the maximum number of retries is exceeded.

PfeIdxRpcResendCount This option specifies the maximum number of IDEX request message retransmissions until the function times out. If no response is received to any of the requests, the driver fails to communicate and raises an error. The minimum value is 1 and means an RPC request will be sent only once. The larger the number the more times the request will be retried.

3.7 Runtime errors

The Ethernet driver generates the following DEM extended production errors at runtime:

3.7.1 Standardized Extended Production Errors

Function	Error code	Condition triggering the error
Eth_MainFunction	ETH_E_ACCESS	Controller access failed
Eth_MainFunction	ETH_E_RX_FRAMES_LOST	Rx frame lost detected
Eth_MainFunction	ETH_E_CRC	CRC failure detected
Eth_MainFunction	ETH_E_UNDERSIZEFRAME	Undersized frame detected
Eth_MainFunction	ETH_E_OVERSIZEFRAME	Oversized frame detected

3.7.2 Vendor-Specific Extended Production Errors

Function	Error code	Condition triggering the error
Eth_MainFunction	ETH_E_DRIVER_RUNTIME_ERR_IP_READY	Timeout when detect IP-ready
Eth_MainFunction	ETH_E_DRIVER_RUNTIME_ERR_INTERNAL	Common Driver Runtime Error
Eth_MainFunction	ETH_E_PFE_ECC_ERR	PFE ECC Error
Eth_MainFunction	ETH_E_PFE_WDT_ERR	PFE Watchdog Error
Eth_MainFunction	ETH_E_EMAC0_ECC_UNCORRECTABLE_ERR	EMAC0 Uncorrectable Error
Eth_MainFunction	ETH_E_EMAC0_ECC_ADDRESS_ERR	EMAC0 Address Error
Eth_MainFunction	ETH_E_EMAC0_PARITY_ERR	EMAC0 Parity Error
Eth_MainFunction	ETH_E_EMAC0_WDT_ERR	EMAC0 Watchdog Error
Eth_MainFunction	ETH_E_EMAC1_ECC_UNCORRECTABLE_ERR	EMAC1 Uncorrectable Error
Eth_MainFunction	ETH_E_EMAC1_ECC_ADDRESS_ERR	EMAC1 Address Error
Eth_MainFunction	ETH_E_EMAC1_PARITY_ERR	EMAC1 Parity Error
Eth_MainFunction	ETH_E_EMAC1_WDT_ERR	EMAC1 Watchdog Error
Eth_MainFunction	ETH_E_EMAC2_ECC_UNCORRECTABLE_ERR	EMAC2 Uncorrectable Error
Eth_MainFunction	ETH_E_EMAC2_ECC_ADDRESS_ERR	EMAC2 Address Error
Eth_MainFunction	ETH_E_EMAC2_PARITY_ERR	EMAC2 Parity Error
Eth_MainFunction	ETH_E_EMAC2_WDT_ERR	EMAC2 Watchdog Error
Eth_MainFunction	ETH_E_BUS_ERR	Bus Error
Eth_MainFunction	ETH_E_PFE_PARITY_ERR	PFE Parity Error
Eth_MainFunction	ETH_E_FAIL_STOP_HW_ERR	FailStop HW Error
Eth_MainFunction	ETH_E_FAIL_STOP_FW_ERR	FailStop FW Error
Eth_MainFunction	ETH_E_FAIL_STOP_SW_ERR	FailStop SW Error
Eth_MainFunction	ETH_E_BMU_FULL	BMU Full
Eth_MainFunction	ETH_E_BMU_ERR	BMU Error
Eth_MainFunction	ETH_E_PE_STALL_ERR	PE Stall Error
Eth_MainFunction	ETH_E_PE_EXCEPTION_ERR	PE Exception Error
Eth_MainFunction	ETH_E_PE_FW_ERR	General Firmware Error
Eth_MainFunction	ETH_E_HIF_ERR	HIF General Error

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [Eth](#)
 - Container [EthConfigSet](#)
 - * Container [EthCtrlConfig](#)
 - Parameter [EthCtrlEnableMii](#)
 - Parameter [EthCtrlEnableRxInterrupt](#)
 - Parameter [EthCtrlEnableTxInterrupt](#)
 - Parameter [EthCtrlIdx](#)
 - Parameter [EthCtrlMacLayerSpeed](#)
 - Parameter [EthCtrlMacLayerSubType](#)
 - Parameter [EthCtrlMacLayerType](#)
 - Parameter [EthCtrlPhyAddress](#)
 - Reference [EthCtrlEcucPartitionRef](#)
 - Container [EthCtrlConfigEgress](#)
 - Reference [EthCtrlConfigEgressLastSchedulerRef](#)
 - Container [EthCtrlConfigEgressFifo](#)
 - Parameter [EthCtrlConfigEgressFifoBufLenByte](#)
 - Parameter [EthCtrlConfigEgressFifoBufTotal](#)
 - Parameter [EthCtrlConfigEgressFifoIdx](#)
 - Parameter [EthCtrlConfigEgressFifoPriorityAssignment](#)
 - Container [EthVendorSpecific](#)
 - Container [EthCtrlConfigScheduler](#)
 - Container [EthCtrlConfigSchedulerPredecessor](#)
 - Parameter [EthCtrlConfigSchedulerPredecessorOrder](#)
 - Reference [EthCtrlConfigSchedulerPredecessorRef](#)
 - Container [EthCtrlConfigSchedulerPredecessorVendorSpecific](#)
 - Parameter [EthCtrlConfigSchedulerBandwidth](#)
 - Container [EthCtrlConfigSchedulerVendorSpecific](#)
 - Parameter [EthCtrlConfigSchedulerAlgorithm](#)
 - Parameter [EthCtrlConfigSchedulerRateMode](#)
 - Parameter [EthCtrlConfigTotalPacketsPerCycle](#)

- Parameter [EthCtrlConfigTotalQuantumPerCycle](#)
- Container [EthCtrlConfigShaper](#)
- Parameter [EthCtrlConfigShaperIdleSlope](#)
- Reference [EthCtrlConfigShaperPredecessorFifoRef](#)
- Container [EthCtrlConfigShaperVendorSpecific](#)
- Parameter [EthCtrlConfigShaperIdx](#)
- Parameter [EthCtrlConfigShaperRateMode](#)
- Parameter [EthCtrlConfigShaperMaxCredit](#)
- Parameter [EthCtrlConfigShaperMinCredit](#)
- Container [EthCtrlConfigIngress](#)
- Container [EthCtrlConfigIngressFifo](#)
- Parameter [EthCtrlConfigIngressFifoBufLenByte](#)
- Parameter [EthCtrlConfigIngressFifoBufTotal](#)
- Parameter [EthCtrlConfigIngressFifoIdx](#)
- Parameter [EthCtrlConfigIngressFifoPriorityAssignment](#)
- Container [EthDemEventParameterRefs](#)
- Reference [ETH_E_ACCESS](#)
- Reference [ETH_E_ALIGNMENT](#)
- Reference [ETH_E_CRC](#)
- Reference [ETH_E_LATECOLLISION](#)
- Reference [ETH_E_MULTIPLECOLLISION](#)
- Reference [ETH_E_OVERSIZEFRAME](#)
- Reference [ETH_E_RX_FRAMES_LOST](#)
- Reference [ETH_E_SINGLECOLLISION](#)
- Reference [ETH_E_UNDERSIZEFRAME](#)
- Container [EthVendorSpecific](#)
- Parameter [EthCtrlHwPort](#)
- Parameter [EthCtrlEthIfIdx](#)
- Parameter [EthDuplexMode](#)
- Parameter [EthEmacLoopEn](#)
- Parameter [EthPromiscuousEn](#)
- Parameter [AcceptAllTraffic](#)
- Parameter [EthEthTrcvDriverVendorIdAndApiInfix](#)
- Parameter [EthEthSwtDriverVendorIdAndApiInfix](#)
- Container [EthGeneral](#)
 - * Parameter [EthDevErrorDetect](#)
 - * Parameter [EthGetClassStatisticApi](#)
 - * Parameter [EthGetCounterValuesApi](#)
 - * Parameter [EthGetDropCountApi](#)
 - * Parameter [EthGetEtherStatsApi](#)
 - * Parameter [EthGetPFESStatisticApi](#)
 - * Parameter [EthGetRxStatsApi](#)
 - * Parameter [EthGetTxErrorCounterValuesApi](#)
 - * Parameter [EthGetTxStatsApi](#)
 - * Parameter [EthGlobalTimeSupport](#)

- * Parameter [EthIndex](#)
- * Parameter [EthMainFunctionPeriod](#)
- * Parameter [EthMaxCtrlsSupported](#)
- * Parameter [EthVersionInfoApi](#)
- * Reference [EthEcucPartitionRef](#)
- * Container [EthCtrlOffloading](#)
 - Parameter [EthCtrlEnableOffloadChecksumICMP](#)
 - Parameter [EthCtrlEnableOffloadChecksumIPv4](#)
 - Parameter [EthCtrlEnableOffloadChecksumTCP](#)
 - Parameter [EthCtrlEnableOffloadChecksumUDP](#)
- * Container [EthVendorSpecific](#)
 - Parameter [EthDisableDemEventDetect](#)
 - Parameter [EthEnableUserModeSupport](#)
 - Parameter [EthEnableClause45API](#)
 - Parameter [EthSwtManagementSupportApi](#)
 - Parameter [EthReleaseTxBufferApi](#)
 - Parameter [EthPfeMultiInstance](#)
 - Parameter [EthPfeSlaveMode](#)
 - Parameter [EthPfeMasterHif](#)
 - Parameter [EthCommonHif](#)
 - Parameter [EthSlaveHifMasterUpTimeout](#)
 - Parameter [SlaveReadStatCounterOfEmacDirectly](#)
 - Parameter [EnERR051211Workaround](#)
 - Parameter [EnableBmuInterrupt](#)
 - Parameter [PfeEnableFCI](#)
 - Parameter [PfeEnableRouting](#)
 - Parameter [PfeRtCollisionSize](#)
 - Parameter [PfeRtHashSize](#)
 - Parameter [PfeConnStatsSize](#)
 - Parameter [PfeEnableBridge](#)
 - Parameter [PfeL2brDomainsMax](#)
 - Parameter [PfeL2brStaticsEnable](#)
 - Parameter [PfeL2brStaticsMax](#)
 - Parameter [VarEthBmu2BufCnt](#)
 - Parameter [MaxNumFwFeaturesSupported](#)
 - Parameter [PfeClassFwLoadFrom](#)
 - Parameter [PfeClassFwMemoryAddress](#)
 - Parameter [PfeClassFwArrayLabel](#)
 - Parameter [FwFeaturesBufferSize](#)
 - Parameter [FwMessagesBufferSize](#)
 - Parameter [PfeFciMaximumFpTableNumber](#)

- Parameter [PfeFciMaximumFpRuleNumber](#)
- Parameter [PfeCfgMacDbEntriesMax](#)
- Parameter [PfeLogIfNameLengthMax](#)
- Parameter [PfeMirrorsMax](#)
- Parameter [PfeChannelBdFlushApi](#)
- Parameter [PfeChannelBdFlushMaxBdCount](#)
- Parameter [PfeChannelBdFlushTimeoutCounter](#)
- Parameter [PfeIdxRpcResendTime](#)
- Parameter [PfeIdxRpcResendCount](#)
- Container [EnableFCIOwnership](#)
- Parameter [PfeEnableFCIOwnershipHIF0](#)
- Parameter [PfeEnableFCIOwnershipHIF1](#)
- Parameter [PfeEnableFCIOwnershipHIF2](#)
- Parameter [PfeEnableFCIOwnershipHIF3](#)
- Container [EthEMAC0Config](#)
- Parameter [EthCtrlEnableMii](#)
- Parameter [EthDuplexMode](#)
- Parameter [EthCtrlMacLayerSpeed](#)
- Parameter [EthCtrlMacLayerSubType](#)
- Parameter [EthCtrlMacLayerType](#)
- Parameter [IEEE1588ClockAttachedToGMAC](#)
- Parameter [IEEE1588ClockOutputFrequencyHz](#)
- Parameter [EnablePpsOutput](#)
- Parameter [PpsOutputPeriod](#)
- Parameter [PpsOutputPulseWidth](#)
- Container [EthEMAC0QueueSizes](#)
- Parameter [EMAC0Q0Size](#)
- Parameter [EMAC0Q1Size](#)
- Parameter [EMAC0Q2Size](#)
- Parameter [EMAC0Q3Size](#)
- Parameter [EMAC0Q4Size](#)
- Parameter [EMAC0Q5Size](#)
- Parameter [EMAC0Q6Size](#)
- Parameter [EMAC0Q7Size](#)
- Container [EthEMAC1Config](#)
- Parameter [EthCtrlEnableMii](#)
- Parameter [EthDuplexMode](#)
- Parameter [EthCtrlMacLayerSpeed](#)
- Parameter [EthCtrlMacLayerSubType](#)
- Parameter [EthCtrlMacLayerType](#)
- Parameter [IEEE1588ClockAttachedToEMAC0](#)

- Parameter [IEEE1588ClockOutputFrequencyHz](#)
- Container [EthEMAC1QueueSizes](#)
- Parameter [EMAC1Q0Size](#)
- Parameter [EMAC1Q1Size](#)
- Parameter [EMAC1Q2Size](#)
- Parameter [EMAC1Q3Size](#)
- Parameter [EMAC1Q4Size](#)
- Parameter [EMAC1Q5Size](#)
- Parameter [EMAC1Q6Size](#)
- Parameter [EMAC1Q7Size](#)
- Container [EthEMAC2Config](#)
- Parameter [EthCtrlEnableMii](#)
- Parameter [EthDuplexMode](#)
- Parameter [EthCtrlMacLayerSpeed](#)
- Parameter [EthCtrlMacLayerSubType](#)
- Parameter [EthCtrlMacLayerType](#)
- Parameter [IEEE1588ClockAttachedToEMAC0](#)
- Parameter [IEEE1588ClockOutputFrequencyHz](#)
- Container [EthEMAC2QueueSizes](#)
- Parameter [EMAC2Q0Size](#)
- Parameter [EMAC2Q1Size](#)
- Parameter [EMAC2Q2Size](#)
- Parameter [EMAC2Q3Size](#)
- Parameter [EMAC2Q4Size](#)
- Parameter [EMAC2Q5Size](#)
- Parameter [EMAC2Q6Size](#)
- Parameter [EMAC2Q7Size](#)
- Container [EthDriverRuntimeErrorsDemEventParameterRefs](#)
- Reference [ETH_E_DRIVER_RUNTIME_ERR_IP_READY](#)
- Reference [ETH_E_DRIVER_RUNTIME_ERR_INTERNAL](#)
- Container [EthPfeEccDemEventParameterRefs](#)
- Reference [ETH_E_PFE_ECC_ERR](#)
- Container [EthPfeWdtDemEventParameterRefs](#)
- Reference [ETH_E_PFE_WDT_ERR](#)
- Container [EthEmacDemEventParameterRefs](#)
- Reference [ETH_E_EMAC0_ECC_UNCORRECTABLE_ERR](#)
- Reference [ETH_E_EMAC0_ECC_ADDRESS_ERR](#)
- Reference [ETH_E_EMAC0_PARITY_ERR](#)
- Reference [ETH_E_EMAC0_WDT_ERR](#)
- Reference [ETH_E_EMAC1_ECC_UNCORRECTABLE_ERR](#)
- Reference [ETH_E_EMAC1_ECC_ADDRESS_ERR](#)

- Reference [ETH_E_EMAC1_PARITY_ERR](#)
- Reference [ETH_E_EMAC1_WDT_ERR](#)
- Reference [ETH_E_EMAC2_ECC_UNCORRECTABLE_ERR](#)
- Reference [ETH_E_EMAC2_ECC_ADDRESS_ERR](#)
- Reference [ETH_E_EMAC2_PARITY_ERR](#)
- Reference [ETH_E_EMAC2_WDT_ERR](#)
- Container [EthBusDemEventParameterRefs](#)
- Reference [ETH_E_BUS_ERR](#)
- Container [EthPfeParityDemEventParameterRefs](#)
- Reference [ETH_E_PFE_PARITY_ERR](#)
- Container [EthFailStopDemEventParameterRefs](#)
- Reference [ETH_E_FAIL_STOP_HW_ERR](#)
- Reference [ETH_E_FAIL_STOP_FW_ERR](#)
- Reference [ETH_E_FAIL_STOP_SW_ERR](#)
- Container [EthBmuDemEventParameterRefs](#)
- Reference [ETH_E_BMU_FULL](#)
- Reference [ETH_E_BMU_ERR](#)
- Container [EthPeDemEventParameterRefs](#)
- Reference [ETH_E_PE_STALL_ERR](#)
- Reference [ETH_E_PE_EXCEPTION_ERR](#)
- Reference [ETH_E_PE_FW_ERR](#)
- Container [EthHifDemEventParameterRefs](#)
- Reference [ETH_E_HIF_ERR](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)
 - * Parameter [ArReleaseRevisionVersion](#)
 - * Parameter [ModuleId](#)
 - * Parameter [SwMajorVersion](#)
 - * Parameter [SwMinorVersion](#)
 - * Parameter [SwPatchVersion](#)
 - * Parameter [VendorApiInfix](#)
 - * Parameter [VendorId](#)

4.1 Module Eth

Configuration of the Eth_43_PFE (Ethernet Driver) module.

Included containers:

- [EthConfigSet](#)
- [EthGeneral](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME, VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container EthConfigSet

This container contains the configuration parameters and sub containers of the AUTOSAR Eth module.

Included subcontainers:

- [EthCtrlConfig](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Container EthCtrlConfig

Configuration of the individual controller

Included subcontainers:

- [EthCtrlConfigEgress](#)
- [EthCtrlConfigIngress](#)
- [EthDemEventParameterRefs](#)
- [EthVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.4 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access. This option is only available on master driver and for EMAC interfaces.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.5 Parameter EthCtrlEnableRxInterrupt

Enables / Disables receive interrupt

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.6 Parameter EthCtrlEnableTxInterrupt

Enables / Disables transmit interrupt

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.7 Parameter EthCtrlIdx

Specifies the instance ID of the configured controller.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.8 Parameter EthCtrlMacLayerSpeed

This bit = 0 & Port Select(Eth line speed) = 1 => Speed = 10Mbps.

This bit = 0 & Port Select = 0 => Speed = 1Gbps.

This bit = 1 & Port Select = 1 => Speed = 100Mbps.

This bit = 1 & Port Select = 0 => Speed = 2.5Gbps.

This option is only available on master driver and for EMAC interfaces.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_1G
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_2500M']

4.9 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of ethernet controller.

This option is only available on master driver and for EMAC interfaces.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	REDUCED
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.10 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

This option is only relevant on master driver and for EMAC interfaces, otherwise it is ignored.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XGMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII']

4.11 Parameter EthCtrlPhyAddress

Specifies the unique 48-bit physical address (MAC address) of the controller in network byte order. Depending on EthCtrlPort option, the address is set to associated EMAC or to AUX interface.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	66:55:44:33:22:11

4.12 Reference EthCtrlEcucPartitionRef

Maps the ETH driver to zero or multiple ECUC partitions to make the driver API available in the according partition. The Eth driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.13 Container EthCtrlConfigEgress

Controller Configuration Egress : Egress Controller configuration of the individual controller

Included subcontainers:

- [EthCtrlConfigEgressFifo](#)
- [EthCtrlConfigScheduler](#)
- [EthCtrlConfigShaper](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.14 Reference EthCtrlConfigEgressLastSchedulerRef

Reference to the scheduler which is the last in the egress structure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔ Egress/EthCtrlConfigScheduler

4.15 Container EthCtrlConfigEgressFifo

Controller Configuration Egress : Egress Controller configuration of the individual controller

Included subcontainers:

- [EthVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.16 Parameter EthCtrlConfigEgressFifoBufLenByte

Limits the maximum transmit buffer length (frame length) in bytes. The minimum value is 64 and maximum 1522. Maximal frame payload is equal to the configured value decreased by 14 bytes that are used for L2 header. Size of

allocated buffers will be increased by 4 bytes for driver's internal metadata and aligned up to 8 bytes. This option does not apply for HIF_NOCOPY interface, which uses BMU buffers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1522
max	1522
min	64

4.17 Parameter EthCtrlConfigEgressFifoBufTotal

Number of transmit buffers. The minimum value is 2 and maximum 512

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4
max	512
min	2

4.18 Parameter EthCtrlConfigEgressFifoIdx

Configures the number of transmit buffers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	254
min	0

4.19 Parameter EthCtrlConfigEgressFifoPriorityAssignment

Message egress priority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.20 Container EthVendorSpecific

Vendor specific: Vendor specific configuration of the individual controller

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.21 Container EthCtrlConfigScheduler

Controller Configuration Scheduler: Represents a Scheduler on the egress side.

Included subcontainers:

- [EthCtrlConfigSchedulerPredecessor](#)
- [EthCtrlConfigSchedulerVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.22 Container EthCtrlConfigSchedulerPredecessor

Controller Configuration Scheduler Predecessors: Defines an ordered list of predecessors for this scheduler.

Included subcontainers:

- [EthCtrlConfigSchedulerPredecessorVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite

Property	Value
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.23 Parameter EthCtrlConfigSchedulerPredecessorOrder

Defines the order of the scheduler predecessors.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.24 Reference EthCtrlConfigSchedulerPredecessorRef

Choice reference to the scheduler predecessor.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigEgressFifo', '/AUTOSAR/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigScheduler', '/AUTOSAR/EcuDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigShaper']

4.25 Container EthCtrlConfigSchedulerPredecessorVendorSpecific

Scheduler Predecessor vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.26 Parameter EthCtrlConfigSchedulerBandwidth

Represents the bandwidth percentage allocated for current scheduler input.

In SCHED_ALGO_RR and SCHED_ALGO_PQ, this parameter is irrelevant.

In SCHED_ALGO_WRR, weight for current scheduler input = (EthTxSchedulerBandwidth / 100) * EthTotalPacketsPerCycle.

In SCHED_ALGO_DWRR, weight for current scheduler input = (EthTxSchedulerBandwidth / 100) * EthTotalQuantumPerCycle.

This field is editable only when the scheduling algorithm is set to SCHED_ALGO_WRR or SCHED_ALGO_DWRR.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	10
max	100
min	1

4.27 Container EthCtrlConfigSchedulerVendorSpecific

Scheduler vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.28 Parameter EthCtrlConfigSchedulerAlgorithm

Configures the Tx Scheduler Algorithm.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	SCHED_ALGO_RR
literals	['SCHED_ALGO_RR', 'SCHED_ALGO_WRR', 'SCHED_ALGO_DWRR', 'SCHED_ALGO_PQ']

4.29 Parameter EthCtrlConfigSchedulerRateMode

Configures the Tx Scheduler rate mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	RATE_MODE_PACKET_RATE
literals	['RATE_MODE_DATA_RATE', 'RATE_MODE_PACKET_RATE']

4.30 Parameter EthCtrlConfigTotalPacketsPerCycle

Configures the total weight value (in packets) per arbitration cycle from which individual scheduler input weights are calculated based on their allocated bandwidths.

For example, if scheduler inputs (I2, I1, I0) are configured with bandwidths (50%, 25%, 25%) and EthCtrlConfigTotalPacketsPerCycle = 4 packets,

then the individual scheduler input weights are $(4 * 0.5, 4 * 0.25, 4 * 0.25) = (2, 1, 1)$ packets.

This field is editable only when the scheduler algorithm is set to SCHED_ALGO_WRR.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	4294967295
min	1

4.31 Parameter EthCtrlConfigTotalQuantumPerCycle

Configures the total quantum value (in bytes) per arbitration cycle from which individual scheduler input quantum values are calculated based on their allocated bandwidths.

For example, if scheduler input (I2, I1, I0) are configured with bandwidths (50%, 25%, 25%) and EthCtrlConfigTotalQuantumPerCycle = 2000 bytes,

then the individual scheduler input quantum values are $(2000 * 0.5, 2000 * 0.25, 2000 * 0.25) = (1000, 500, 500)$ bytes.

This field is editable only when the scheduler algorithm is set to SCHED_ALGO_DWRR.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	1

4.32 Container EthCtrlConfigShaper

Controller Configuration Shaper: Represents a Shaper at the egress side.

Included subcontainers:

- [EthCtrlConfigShaperVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.33 Parameter EthCtrlConfigShaperIdleSlope

Defines the increase of credit in bits per second for the AVB shaper.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	2500000000
min	0

4.34 Reference EthCtrlConfigShaperPredecessorFifoRef

Reference to the fifo which is the predecessor for this shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↔Egress/EthCtrlConfigEgressFifo

4.35 Container EthCtrlConfigShaperVendorSpecific

Shaper vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.36 Parameter EthCtrlConfigShaperIdx

Specifies the shaper index of the configured controller.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	true
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	3
min	0

4.37 Parameter EthCtrlConfigShaperRateMode

Configures the Shaper rate mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	RATE_MODE_PACKET_RATE
literals	['RATE_MODE_DATA_RATE', 'RATE_MODE_PACKET_RATE']

4.38 Parameter EthCtrlConfigShaperMaxCredit

Defines the maximum value (in bits) that can be accumulated in the credit parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100
max	4194303
min	0

4.39 Parameter EthCtrlConfigShaperMinCredit

Defines the minimum value (in bits) that can be accumulated in the credit parameter.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100
max	4194303
min	0

4.40 Container EthCtrlConfigIngress

Controller Configuration Ingress : Configuration of one Ethernet controller ingress behavior.

Included subcontainers:

- [EthCtrlConfigIngressFifo](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.41 Container EthCtrlConfigIngressFifo

Controller Configuration Ingress : Represents a Fifo at the ingress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.42 Parameter EthCtrlConfigIngressFifoBufLenByte

Only value 1522 is supported. Limits the maximum receive buffer length (frame length) in bytes. Maximum frame payload is equal to the configured value decreased by 14 bytes that are used for the frame header. Size of allocated buffers will be increased by 16 bytes for internal Rx header and aligned up to 8 bytes. Rx buffer size is configurable per HIF interface, so if this values for all controllers and their FIFOs are not the same, maximum of them will be used. This option does not apply for HIF_NOCOPY interface, which uses BMU buffers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1522
max	1522
min	1522

4.43 Parameter EthCtrlConfigIngressFifoBufTotal

Number of receive buffers. The minimum value is 2 and maximum 512. Rx buffers are used also for delivery of PTP timestamps from PFE, which should be considered when choosing this value. Since single HIF is used, all Rx buffers from all controllers/queues are put together into single buffer pool, which is extended by 4 buffers if 'Multiple PFE drivers support (master-slave)' option is enabled.

For HIF_NOCPY interface, only buffer descriptors are allocated based on this option, no buffers. Buffers from BMU2 are used instead (see option VarEthBmu2BufCnt).

For standard HIF, the total number of allocated buffers is automatically extended by 72 additional buffers for Errata ERR051211 workaround, unless disabled.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4
max	512
min	2

4.44 Parameter EthCtrlConfigIngressFifoIdx

Ingress Fifo index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	0
min	0

4.45 Parameter EthCtrlConfigIngressFifoPriorityAssignment

Message ingress priority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.46 Container EthDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced

DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.47 Reference ETH_E_ACCESS

Reference to the DemEventParameter which shall be issued when the error "Controller access failed" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.48 Reference ETH_E_ALIGNMENT

Reference to the DemEventParameter which shall be issued when the error "Alignment Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.49 Reference ETH_E_CRC

Reference to the DemEventParameter which shall be issued when the error "CRC Failure" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.50 Reference ETH_E_LATECOLLISION

Reference to the DemEventParameter which shall be issued when the error "Late Collisions" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.51 Reference ETH_E_MULTIPLECOLLISION

Reference to the DemEventParameter which shall be issued when the error "Multiple Collisions" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.52 Reference ETH_E_OVERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error "Oversized Frame" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.53 Reference ETH_E_RX_FRAMES_LOST

Reference to the DemEventParameter which shall be issued when the error "receive frames lost" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.54 Reference ETH_E_SINGLECOLLISION

Reference to the DemEventParameter which shall be issued when the error "Single Collisions" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.55 Reference ETH_E_UNDERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error "Undersized Frame" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.56 Container EthVendorSpecific

Vendor specific: Vendor specific configuration of the individual controller

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.57 Parameter EthCtrlHwPort

Vendor specific: Defines which interface (MAC, AUX or HIF) the ethernet controller shall be attach to.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	EMAC0
literals	['EMAC0', 'EMAC1', 'EMAC2', 'AUX', 'HIF0', 'HIF1', 'HIF2', 'HIF3', 'HIF_↔ NOCOPY']

4.58 Parameter EthCtrlEthIfIdx

Vendor specific: Defines the index of the controller in context of EthIf driver. This manual assignemnt of this value is only temporary workaround before proper solution is implemented.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.59 Parameter EthDuplexMode

Defines the controller's duplex mode. Only available for EMAC controllers.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.60 Parameter EthEmacLoopEn

Vendor specific: Enables / Disables Loopback Mode on associated EMAC. This mode is intended for testing. Only available for EMAC controllers.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.61 Parameter EthPromiscuousEn

Vendor specific: Enables / Disables Promiscuous Mode. In current version this option is not available for slave drivers (see 'Slave driver' option). Only available for EMAC controllers.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.62 Parameter AcceptAllTraffic

This parameter is for the case where the HIF associated to this instance and the EMAC corresponding to this controller are part of a bridge domain. If enabled, all ingress traffic received on the corresponding EMAC will be received by this controller. If disabled, only management traffic from the EMAC will be received by this controller. All other traffic will be received on the AUX controller. It is recommended to disable this parameter when the corresponding EMAC is part of bridge domain. Not available on HIF0, HIF1, HIF2, HIF3 and HIF_NOCPY interfaces.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: POST-BUILD
default Value	true

4.63 Parameter EthEthTrcvDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthTrcv driver, if any.

This is used to include the correct header file and to point to the EthTrcv callbacks.

Example: 43_PHYGROUPB for a 43_PHYGROUPB driver or leave it blank to include EthTrcv.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	

4.64 Parameter EthEthSwtDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthSwt driver, if any.

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This is used to include the correct header file and to point to the EthSwt APIs.

Example: 43_SJA1110 for a 43_SJA1110 driver or leave it blank to include EthSwt.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	

4.65 Container EthGeneral

General configuration of Ethernet Driver module

Included subcontainers:

- [EthCtrlOffloading](#)
- [EthVendorSpecific](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.66 Parameter EthDevErrorDetect

Enables / Disables development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.67 Parameter EthGetClassStatisticApi

Enables / Disables Eth_43_PFE_GetClassStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.68 Parameter EthGetCounterValuesApi

Enables / Disables Eth_GetCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.69 Parameter EthGetDropCountApi

Obsolete and kept only for backwards compatibility. The Eth_GetDropCount API has been replaced with Eth_GetCounterValues since R4.3.1. Use parameter EthGetCounterValuesApi instead.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.70 Parameter EthGetEtherStatsApi

Obsolete and kept only for backwards compatibility. The Eth_GetEtherStats API has been replaced with the Eth_GetTxStats and Eth_GetRxStats APIs since R4.3.1.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.71 Parameter EthGetPFESTatisticApi

Enables / Disables Eth_43_PFE_GetPFESTats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.72 Parameter EthGetRxStatsApi

Enables / Disables Eth_GetRxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.73 Parameter EthGetTxErrorCounterValuesApi

Enables / Disables Eth_GetTxErrorCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.74 Parameter EthGetTxStatsApi

Enables / Disables Eth_GetTxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
	VARIANT-LINK-TIME: LINK
valueConfigClasses	

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.75 Parameter EthGlobalTimeSupport

Enables/Disables the GlobalTime APIs used amongst others by Global Time Synchronization over Ethernet.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.76 Parameter EthIndex

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0
max	255
min	0

4.77 Parameter EthMainFunctionPeriod

Specifies the period of main function Eth_MainFunction in seconds.
 mation is required by BSW scheduler and can be used also in the driver.
 define Infinity as limitation, but it is abnormal if the period is longer than 65.535 second

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ASR

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0.1
max	1.0
min	0.001

4.78 Parameter EthMaxCtrlsSupported

Limits the total number of supported controllers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	3
max	8
min	1

4.79 Parameter EthVersionInfoApi

Enables / Disables version info API

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.80 Reference EthEcucPartitionRef

Maps the Eth driver to zero or one ECUC partition to make the modules API available in this partition.

EN: Tags: atp.Status=draft

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.81 Container EthCtrlOffloading

Configuration of hardware offloading features.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.82 Parameter EthCtrlEnableOffloadChecksumICMP

Enables / Disables hardware offloading for ICMP checksums.

If disabled, the FW will be configured to enable receive_malformed. This has the side effect of allowing frames with invalid Ethernet CRC to be received by the driver

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.83 Parameter EthCtrlEnableOffloadChecksumIPv4

Enables / Disables hardware offloading for IPv4 checksums.

If disabled, the FW will be configured to enable receive_malformed. This has the side effect of allowing frames with invalid Ethernet CRC to be received by the driver

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.84 Parameter EthCtrlEnableOffloadChecksumTCP

Enables / Disables hardware offloading for TCP checksums.

If disabled, the FW will be configured to enable receive_malformed. This has the side effect of allowing frames with invalid Ethernet CRC to be received by the driver

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.85 Parameter EthCtrlEnableOffloadChecksumUDP

Enables / Disables hardware offloading for UDP checksums.

If disabled, the FW will be configured to enable receive_malformed. This has the side effect of allowing frames with invalid Ethernet CRC to be received by the driver

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.86 Container EthVendorSpecific

Vendor specific: General vendor specific configuration of Ethernet Driver module

Included subcontainers:

- [EnableFCIOwnership](#)
- [EthEMAC0Config](#)
- [EthEMAC1Config](#)
- [EthEMAC2Config](#)
- [EthDriverRuntimeErrorsDemEventParameterRefs](#)
- [EthPfeEccDemEventParameterRefs](#)
- [EthPfeWdtDemEventParameterRefs](#)
- [EthEmacDemEventParameterRefs](#)
- [EthBusDemEventParameterRefs](#)
- [EthPfeParityDemEventParameterRefs](#)
- [EthFailStopDemEventParameterRefs](#)
- [EthBmuDemEventParameterRefs](#)
- [EthPeDemEventParameterRefs](#)
- [EthHifDemEventParameterRefs](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.87 Parameter EthDisableDemEventDetect

Vendor specific: When this option is ON the Diagnostic Event Manager is disabled, no diagnostic error events are reported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.88 Parameter EthEnableUserModeSupport

When this field is enabled, the Eth_43_PFE module will be run in the User Mode, therefore, the protected registers for the module Eth_43_PFE can still be accessed

normally, except the registers can only be configured to the supervisor mode. When it comes to function implementing write to protected registers which don't exist in the

Eth_43_PFE, so these can be called using the function named `trusted_callparams()`.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.89 Parameter EthEnableClause45API

Vendor specific: When ON then API to access Ethernet transceiver using Clause 45 will be available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.90 Parameter EthSwtManagementSupportApi

Enables / Disables Switch Management APIs.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.91 Parameter EthReleaseTxBufferApi

Enables / Disables Eth_43_PFE_ReleaseTxBuffer API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.92 Parameter EthPfeMultiInstance

Vendor specific: When ON then role of this driver can be selected: master or slave and other driver(s) (Linux/QNX/another MCAL driver) can be used on the same system simultaneously. In current version this option is not available if option 'Use multiple HIF interfaces' is set.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.93 Parameter EthPfeSlaveMode

Vendor specific: This option depends on option "Multiple PFE drivers support (master-slave)" being enabled.

When ON then this driver requires another driver to be simultaneously running on the device. The other driver shall be configured as master.

When OFF then this driver is a master driver that can either run alone or it can support one or more slave drivers running on the same device.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.94 Parameter EthPfeMasterHif

Vendor specific: Specifies HIF interface used by master PFE driver. This option is valid only if option 'Slave driver' is set.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	HIF0
literals	['HIF0', 'HIF1', 'HIF2', 'HIF3', 'HIF_NOCPY']

4.95 Parameter EthCommonHif

Vendor specific: Defines the HIF interface which this driver will use.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	HIF0
literals	['HIF0', 'HIF1', 'HIF2', 'HIF3', 'HIF_NOCPY']

4.96 Parameter EthSlaveHifMasterUpTimeout

Timeout value (in milliseconds) used to wait for Master to be started before Slave sends RPC requests. Special value 'zero milliseconds' (default) will ensure that slave will wait for the master indefinitely.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	100000
min	0

4.97 Parameter SlaveReadStatCounterOfEmacDirectly

Vendor specific: This option depends on option "Slave driver" being enabled.

When ON then slave reads statistic counters directly from EMAC registers.

When OFF then slave reads statistic counters from EMAC registers through RPC

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.98 Parameter EnERR051211Workaround

Turn OFF to disable errata ERR051211 workaround. Errata workaround will be automatically disabled if 'multiple PFE drivers' is disabled. In case of multiple PFE drivers running on the SOC, this option must be configured same way in all of them.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.99 Parameter EnableBmuInterrupt

Turn ON to enable BMU Interrupt

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.100 Parameter PfeEnableFCI

Vendor specific: Turn ON to make the FCI API, Routing Table and L2 Bridge available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.101 Parameter PfeEnableRouting

Vendor specific: Turn ON to enable routing.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.102 Parameter PfeRtCollisionSize

Vendor specific: Number of L3 routing table entries available in collision space. collision space is used to form linked lists of entries for the same hash. Default value shall be either 8, 16, 32, 64, 128, 256, 512, 1024, 2048 or 4096.

The value is 256. The

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	256
max	4096
min	8

4.103 Parameter PfeRtHashSize

Vendor specific: Number of L3 routing table entries available in hash space.

This

number also represents number of bits of the hash calculated from the IP header of which is being used to address the entries within the hash space. Default value shall be either 8, 16, 32, 64, 128, 256, 512, 1024, 2048, or 4096.

packets is 256. The value

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	256
max	4096
min	8

4.104 Parameter PfeConnStatsSize

Vendor specific: Number of stats table entries for conntracks. Default

value is 20.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	20
max	4096
min	0

4.105 Parameter PfeEnableBridge

Vendor specific: Turn ON to enable bridge mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.106 Parameter PfeL2brDomainsMax

Maximum number of L2BR domains.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	64
min	2

4.107 Parameter PfeL2brStaticsEnable

Vendor specific: This option determines if L2 bridge static entries can be created using FPP_CMD_L2_STATIC_ENT command.

When ON then allows to create entries up to amount specified in PfeL2brStaticsMax.

When OFF then no entries will be allowed to create.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.108 Parameter PfeL2brStaticsMax

Maximum number of L2 bridge static entries the FCI is allowed to create using FPP_CMD_L2_STATIC_ENT command.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16

Property	Value
max	256
min	1

4.109 Parameter VarEthBmu2BufCnt

Vendor specific: Number of 2048 byte buffers that are used for 2 purposes: 1) for HIF_NOCOPY interface buffers, both Rx and Tx. Even if the NOCOPY interface is used in another driver instance, the buffers are still allocated from here, and 2) by PFE hardware internally for processing Ethernet packets among all HIF and EMAC interfaces. Higher number of buffers allows to better cover bursts of Ethernet traffic. When the option is too low, packets may be dropped during bursts due to buffer depletion.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	512
max	1024
min	32

4.110 Parameter MaxNumFwFeaturesSupported

Vendor specific: The maximum number of FW features.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	22
max	255
min	12

4.111 Parameter PfeClassFwLoadFrom

Vendor specific: Parameter specifies PFE firmware origin.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	PfeClassFwArrayLabel
literals	['PfeClassFwArrayLabel', 'PfeClassFwMemoryAddress']

4.112 Parameter PfeClassFwMemoryAddress

Vendor specific: Parameter specifies hexadecimal PFE firmware storage memory address.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false

Property	Value
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	341F0000

4.113 Parameter PfcClassFwArrayLabel

Vendor specific: Parameter specifies PFE firmware storage array variable name defined as `const uint8 string_entered_into_PfcClassFwArrayLabel_field[];`

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	pfe_class_fw_binary

4.114 Parameter FwFeaturesBufferSize

Vendor specific: The maximum size of class firmware .features ELF file section in bytes. Used to prepare enough buffer for copy of section.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	3000
max	9999
min	0

4.115 Parameter FwMessagesBufferSize

Vendor specific: The maximum size of class firmware .messages ELF file section in bytes. Used to prepare enough buffer for copy of section.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4000
max	9999
min	0

4.116 Parameter PfeFciMaximumFpTableNumber

Maximum number of flexible parser tables that can be registered through FCI command FPP_CMD_FP_TABLE.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	64
min	1

4.117 Parameter PfeFciMaximumFpRuleNumber

Maximum number of flexible parser rules that can be registered through FCI command FPP_CMD_FP_RULE.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	20
max	256
min	1

4.118 Parameter PfeCfgMacDbEntriesMax

Maximum number of MAC entries per interface.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	8
max	64
min	1

4.119 Parameter PfcLogIfNameLengthMax

Maximum length of logical interface name (including \0 termination character).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	32
min	2

4.120 Parameter PfcMirrorsMax

Maximum number of mirrors that can be registered through FCI command FPP_CMD_MIRROR.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	8
max	40
min	0

4.121 Parameter PfeChannelBdFlushApi

Vendor specific: Enable API to perform cached RX BD flush on any channel.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.122 Parameter PfeChannelBdFlushMaxBdCount

Maximum number of slave cached BDs that can be released in a single API call. Set to 0 to disable the limit.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	256
min	0

4.123 Parameter PfeChannelBdFlushTimeoutCounter

Timeout value (in loop turnaround count) used to wait for the packet to be received on the slave HIF when BD flush.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	200
max	4294967295
min	0

4.124 Parameter PfeIdxRpcResendTime

IDEX request timeout in milliseconds between resending IDEX RPC requests.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100
max	65535
min	0

4.125 Parameter PfeIdxRpcResendCount

IDEX request maximum resend count. 1 means only one RPC request will be sent, without further attempts.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	40
max	65535
min	1

4.126 Container EnableFCIOwnership

Slave driver on HIFn can be FCI owner

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.127 Parameter PfeEnableFCIownershipHIF0

Vendor specific: Turn ON to allow slave drivers on HIF0 to become FCI owners.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.128 Parameter PfeEnableFCIownershipHIF1

Vendor specific: Turn ON to allow slave drivers on HIF1 to become FCI owners.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.129 Parameter PfeEnableFCIownershipHIF2

Vendor specific: Turn ON to allow slave drivers on HIF2 to become FCI owners.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.130 Parameter PfeEnableFCIownershipHIF3

Vendor specific: Turn ON to allow slave drivers on HIF3 to become FCI owners.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.131 Container EthEMAC0Config

Configuration for EMAC0

Included subcontainers:

- [EthEMAC0QueueSizes](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.132 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.133 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.134 Parameter EthCtrlMacLayerSpeed

This bit = 0 & Port Select(Eth line speed) = 1 => Speed = 10Mbps.

This bit = 0 & Port Select = 0 => Speed = 1Gbps.

This bit = 1 & Port Select = 1 => Speed = 100Mbps.

This bit = 1 & Port Select = 0 => Speed = 2.5Gbps.

This option is only available on master driver and for EMAC interfaces.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_1G
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_2500M']

4.135 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of ethernet controller.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	REDUCED
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.136 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XGMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII']

4.137 Parameter IEEE1588ClockAttachedToGMAC

Enables external clock for EMAC IEEE1588 timestamping unit. Clock will be attached to GMAC

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.138 Parameter IEEE1588ClockOutputFrequencyHz

Internal clock for timestamping unit. Can be used only when IEEE1588ClockAttachedToGMAC is False

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100000000
max	199999999
min	3906251

4.139 Parameter EnablePpsOutput

Enable PPS output signal generation

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.140 Parameter PpsOutputPeriod

PPS output period [units of PTP reference clock ticks].

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	200000000
max	4294967295
min	2

4.141 Parameter PpsOutputPulseWidth

PPS output pulse width [units of PTP reference clock ticks].

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	100000000
max	4294967295
min	2

4.142 Container EthEMAC0QueueSizes

To set the sizes of TMU queues Q0 - Q7, total maximum size is 255 bytes

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.143 Parameter EMAC0Q0Size

Size of TMU queue 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.144 Parameter EMAC0Q1Size

Size of TMU queue 1.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.145 Parameter EMAC0Q2Size

Size of TMU queue 2.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.146 Parameter EMAC0Q3Size

Size of TMU queue 3.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.147 Parameter EMAC0Q4Size

Size of TMU queue 4.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.148 Parameter EMAC0Q5Size

Size of TMU queue 5.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.149 Parameter EMAC0Q6Size

Size of TMU queue 6.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.150 Parameter EMAC0Q7Size

Size of TMU queue 7.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.151 Container EthEMAC1Config

Configuration for EMAC1

Included subcontainers:

- [EthEMAC1QueueSizes](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.152 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.153 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.154 Parameter EthCtrlMacLayerSpeed

This bit = 0 & Port Select(Eth line speed) = 1 => Speed = 10Mbps.

This bit = 0 & Port Select = 0 => Speed = 1Gbps.

This bit = 1 & Port Select = 1 => Speed = 100Mbps.

This bit = 1 & Port Select = 0 => Speed = 2.5Gbps.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_1G
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_2500M']

4.155 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of ethernet controller.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	REDUCED
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.156 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XGMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII']

4.157 Parameter IEEE1588ClockAttachedToEMAC0

Enables external clock for EMAC IEEE1588 timestamping unit. Clock will be attached to EMAC0

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.158 Parameter IEEE1588ClockOutputFrequencyHz

Internal clock for timestamping unit. Can be used only when IEEE1588ClockAttachedToEMAC0 is False

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100000000
max	199999999
min	3906251

4.159 Container EthEMAC1QueueSizes

To set the sizes of TMU queues Q0 - Q7, total maximum size is 255 bytes

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.160 Parameter EMAC1Q0Size

Size of TMU queue 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.161 Parameter EMAC1Q1Size

Size of TMU queue 1.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.162 Parameter EMAC1Q2Size

Size of TMU queue 2.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.163 Parameter EMAC1Q3Size

Size of TMU queue 3.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.164 Parameter EMAC1Q4Size

Size of TMU queue 4.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.165 Parameter EMAC1Q5Size

Size of TMU queue 5.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.166 Parameter EMAC1Q6Size

Size of TMU queue 6.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.167 Parameter EMAC1Q7Size

Size of TMU queue 7.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.168 Container EthEMAC2Config

Configuration for EMAC2

Included subcontainers:

- [EthEMAC2QueueSizes](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.169 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.170 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.171 Parameter EthCtrlMacLayerSpeed

This bit = 0 & Port Select(Eth line speed) = 1 => Speed = 10Mbps.

This bit = 0 & Port Select = 0 => Speed = 1Gbps.

This bit = 1 & Port Select = 1 => Speed = 100Mbps.

This bit = 1 & Port Select = 0 => Speed = 2.5Gbps.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_1G
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_2500M']

4.172 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of ethernet controller.



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This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	REDUCED
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.173 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

This option is only available on master driver and for EMAC interfaces

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XGMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGMII']

4.174 Parameter IEEE1588ClockAttachedToEMAC0

Enables external clock for EMAC IEEE1588 timestamping unit. Clock will be attached to EMAC0

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.175 Parameter IEEE1588ClockOutputFrequencyHz

Internal clock for timestamping unit. Can be used only when IEEE1588ClockAttachedToEMAC0 is False

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	100000000
max	199999999
min	3906251

4.176 Container EthEMAC2QueueSizes

To set the sizes of TMU queues Q0 - Q7, total maximum size is 255 bytes

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.177 Parameter EMAC2Q0Size

Size of TMU queue 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.178 Parameter EMAC2Q1Size

Size of TMU queue 1.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.179 Parameter EMAC2Q2Size

Size of TMU queue 2.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.180 Parameter EMAC2Q3Size

Size of TMU queue 3.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.181 Parameter EMAC2Q4Size

Size of TMU queue 4.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.182 Parameter EMAC2Q5Size

Size of TMU queue 5.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.183 Parameter EMAC2Q6Size

Size of TMU queue 6.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.184 Parameter EMAC2Q7Size

Size of TMU queue 7.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	31
max	255
min	0

4.185 Container EthDriverRuntimeErrorsDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.186 Reference ETH_E_DRIVER_RUNTIME_ERR_IP_READY

Reference to the DemEventParameter which shall be issued when the error "Timeout when detect IP-ready" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.187 Reference ETH_E_DRIVER_RUNTIME_ERR_INTERNAL

Reference to the DemEventParameter which shall be issued when the common driver runtime errors have occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.188 Container EthPfeEccDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.189 Reference ETH_E_PFE_ECC_ERR

Reference to the DemEventParameter which shall be issued when the error "PFE ECC Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.190 Container EthPfeWdtDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.191 Reference ETH_E_PFE_WDT_ERR

Reference to the DemEventParameter which shall be issued when the error "PFE Watchdog Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.192 Container EthEmacDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.193 Reference

ETH_E_EMAC0_ECC_UNCORRECTABLE_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC0 ECC Uncorrectable Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.194 Reference ETH_E_EMAC0_ECC_ADDRESS_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC0 ECC Address Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.195 Reference ETH_E_EMAC0_PARITY_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC0 Parity Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.196 Reference ETH_E_EMAC0_WDT_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC0 Watchdog Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.197 Reference

ETH_E_EMAC1_ECC_UNCORRECTABLE_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC1 Uncorrectable Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.198 Reference ETH_E_EMAC1_ECC_ADDRESS_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC1 Address Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.199 Reference ETH_E_EMAC1_PARITY_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC1 Parity Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.200 Reference ETH_E_EMAC1_WDT_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC1 Watchdog Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.201 Reference

ETH_E_EMAC2_ECC_UNCORRECTABLE_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC2 Uncorrectable Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.202 Reference ETH_E_EMAC2_ECC_ADDRESS_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC2 Address Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.203 Reference ETH_E_EMAC2_PARITY_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC2 Parity Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.204 Reference ETH_E_EMAC2_WDT_ERR

Reference to the DemEventParameter which shall be issued when the error "EMAC2 Watchdog Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.205 Container EthBusDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.206 Reference ETH_E_BUS_ERR

Reference to the DemEventParameter which shall be issued when the error "Bus Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.207 Container EthPfeParityDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.208 Reference ETH_E_PFE_PARITY_ERR

Reference to the DemEventParameter which shall be issued when the error "PFE Parity Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.209 Container EthFailStopDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.210 Reference ETH_E_FAIL_STOP_HW_ERR

Reference to the DemEventParameter which shall be issued when the error "FailStop HW Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.211 Reference ETH_E_FAIL_STOP_FW_ERR

Reference to the DemEventParameter which shall be issued when the error "FailStop FW Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.212 Reference ETH_E_FAIL_STOP_SW_ERR

Reference to the DemEventParameter which shall be issued when the error "FailStop SW Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.213 Container EthBmuDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.214 Reference ETH_E_BMU_FULL

Reference to the DemEventParameter which shall be issued when the error "BMU Full" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.215 Reference ETH_E_BMU_ERR

Reference to the DemEventParameter which shall be issued when the error "BMU Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.216 Container EthPeDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.217 Reference ETH_E_PE_STALL_ERR

Reference to the DemEventParameter which shall be issued when the error "PE Stall Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.218 Reference ETH_E_PE_EXCEPTION_ERR

Reference to the DemEventParameter which shall be issued when the error "PE Exception Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP

Property	Value
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.219 Reference ETH_E_PE_FW_ERR

Reference to the DemEventParameter which shall be issued when the error "PE FW Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.220 Container EthHifDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.221 Reference ETH_E_HIF_ERR

Reference to the DemEventParameter which shall be issued when the error "HIF General Error" has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.222 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.223 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.224 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.225 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.226 Parameter ModuleId

Module ID of this module from Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	88
max	88
min	88

4.227 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	1
max	1
min	1

4.228 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.229 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.230 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU,

BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name.

In total, the implementation specific name is generated as follows:

[ModuleName]_[VendorId]_[VendorApiInfix][Api name from SWS].

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a

VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1.

It shall not be used for modules with upper multiplicity =1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	PFE

4.231 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43



Chapter 5

Module Index

5.1 Software Specification

Here is a list of all modules:

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Chapter 6

Module Documentation

6.1 Ethernet Driver

6.1.1 Detailed Description

6.1.1.1 Running multiple PFE drivers

Master-Slave mechanism is supported among PFE drivers for Linux, QNX and AUTOSAR (this driver). This allows to simultaneously run multiple drivers on one chip.

Ethernet driver for AUTOSAR can be configured either in Master or Slave mode. Master driver needs to be initialized before Slave drivers can be configured/initialized.

In Slave mode, some features are not available (in current version):

- using promiscuous mode (vendor specific option),
- configuring PHY interfaces (type and speed)

To configure a Slave MCAL driver:

- enable 'Multiple PFE drivers support (Master-Slave)',
- enable 'Slave driver',
- configure 'Master HIF interface' according to Master driver configuration.

To configure Master MCAL driver:

- enable 'Multiple PFE drivers support (Master-Slave)',
- disable 'Slave driver',
- configure 'Master HIF interface' to be the same as 'Common HIF interface',

Handling received Ethernet frames in case an EMAC is used directly by both Master and Slave driver:

- received frames with destination MAC address matching the MAC address of Slave driver are received by Slave driver,
- if reception of broadcast frames is enabled in Slave driver, then all broadcast frames are also received by Slave driver (vendor specific option `EthReceiveBroadcast`),
- if Slave driver adds one or more multicast addresses through function `Eth_43_PFE_UpdatePhysAddrFilter`, then frames with destination addresses matching the filter, are also received by Slave driver.
- all other frames are received by Master driver. This includes also multicast frames that are accepted due to inexact match. For more details refer to [Eth_43_PFE_UpdatePhysAddrFilter](#).

Note

In use cases, where multiple drivers need to use the same EMAC or EMACs, it is recommended to configure some form of routing or bridging among the EMAC(s) and HIFs, to avoid using EMACs directly. For more information refer to section [Driver usage and configuration tips](#)

6.1.1.2 Expected function calls for initialization of Ethernet driver running in Master mode

- `Eth_43_PFE_PreInit` - This function must be called before PFE clock is initialized
- `Eth_43_PFE_Init`

6.1.1.3 Expected function calls for transmission in polling mode

- `Eth_43_PFE_ProvideTxBuffer`
- `Eth_43_PFE_Transmit`
- `Eth_43_PFE_TxConfirmation`
- `Eth_43_PFE_ReleaseTxBuffer`

Function Reference

- [Eth_43_PFE_HifIrqHdlr_0 \(\)](#)
Interrupt handler for HIF0 channel IRQ.
- [Eth_43_PFE_HifIrqHdlr_1 \(\)](#)
Interrupt handler for HIF1 channel IRQ.
- [Eth_43_PFE_HifIrqHdlr_2 \(\)](#)
Interrupt handler for HIF2 channel IRQ.
- [Eth_43_PFE_HifIrqHdlr_3 \(\)](#)
Interrupt handler for HIF3 channel IRQ.
- [Eth_43_PFE_HifNoCpyIrqHdlr \(\)](#)
Interrupt handler for HIF Nocopy channel IRQ.
- [Eth_43_PFE_BmuIrqHdlr \(\)](#)

Interrupt handler for BMUs.

- Std_ReturnType [Eth_43_PFE_Transmit](#) (uint8 u8CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8 *PhysAddrPtr)

Triggers transmission of a previously filled transmit buffer.

- BufReq_ReturnType [Eth_43_PFE_ProvideTxBuffer](#) (uint8 u8CtrlIdx, uint8 Priority, Eth_BufIdxType *BufIdxPtr, uint8 **BufPtr, uint16 *LenBytePtr)

Provides access to a transmit buffer of the specified controller.

- BufReq_ReturnType [Eth_43_PFE_ReleaseTxBuffer](#) (uint8 u8CtrlIdx, Eth_BufIdxType BufIdx)

Release a previously granted TX buffer. The transmission must not have been triggered.

- void [Eth_43_PFE_TxConfirmation](#) (uint8 u8CtrlIdx)

Triggers frame transmission confirmations.

- void [Eth_43_PFE_PreInit](#) (const Eth_43_PFE_ConfigType *CfgPtr)

The function should be used to initialize pfe interfaces.

- void [Eth_43_PFE_Init](#) (const Eth_43_PFE_ConfigType *CfgPtr)

Initializes the Ethernet Driver.

- void [Eth_43_PFE_DeInit](#) (void)

Shutdown the driver gracefully.

- Std_ReturnType [Eth_43_PFE_SetControllerMode](#) (uint8 u8CtrlIdx, Eth_ModeType CtrlMode)

Enables or disables the given controller.

- Std_ReturnType [Eth_43_PFE_GetControllerMode](#) (uint8 u8CtrlIdx, Eth_ModeType *CtrlModePtr)

Obtains the mode of the given controller.

- void [Eth_43_PFE_GetPhysAddr](#) (uint8 u8CtrlIdx, uint8 *PhysAddrPtr)

Obtains the physical source address used by the indexed controller (the node MAC address).

- Std_ReturnType [Eth_43_PFE_UpdatePhysAddrFilter](#) (uint8 u8CtrlIdx, const uint8 *PhysAddrPtr, Eth_FilterActionType Action)

Adds or removes the specified PhysAddrPtr address to or from a multicast address pool in the controller specified by CtrlIdx.

- void [Eth_43_PFE_Receive](#) (uint8 u8CtrlIdx, uint8 FifoIdx, Eth_RxStatusType *RxStatusPtr)

Triggers frames reception notifications.

- Std_ReturnType [Eth_43_PFE_WriteMii](#) (uint8 u8CtrlIdx, uint8 u8TrcvIdx, uint8 u8RegIdx, uint16 u16RegVal)

Write specified transceiver register through the MII.

- Std_ReturnType [Eth_43_PFE_ReadMii](#) (uint8 u8CtrlIdx, uint8 u8TrcvIdx, uint8 u8RegIdx, uint16 *pu16RegValPtr)

Read the specified transceiver register through the MII.

- Std_ReturnType [Eth_43_PFE_WriteMii45](#) (uint8 u8CtrlIdx, uint8 u8TrcvIdx, uint8 u8DevIdx, uint16 u16RegIdx, uint16 u16RegVal)

Write specified transceiver register through the MII (Clause 45)

- Std_ReturnType [Eth_43_PFE_ReadMii45](#) (uint8 u8CtrlIdx, uint8 u8TrcvIdx, uint8 u8DevIdx, uint16 u16RegIdx, uint16 *pu16RegValPtr)

Read the specified transceiver register through the MII (Clause 45)

- Std_ReturnType [Eth_43_PFE_GetClassStats](#) (pfe_ct_classify_stats_t *stat)

Get class statistics from firmware.

- Std_ReturnType [Eth_43_PFE_GetRtableStats](#) (pfe_ct_conntrack_stats_t *stat, uint8 conntrack_index)

Get rtable statistics from firmware.

- Std_ReturnType [Eth_43_PFE_GetL2BridgeDomainStats](#) (pfe_ct_vlan_stats_t *stat, uint32 index_vlan)

Get l2 bridge statistics from firmware.

- Std_ReturnType [Eth_43_PFE_GetL2BridgeStats](#) (Eth_43_PFE_L2BridgeStatsType *stat, uint32 index↵_entry)
Get l2 bridge statistics from firmware.
- Std_ReturnType [Eth_43_PFE_GetWdtStats](#) (Eth_43_PFE_WdtStatsType *stat)
Get wdt statistics from firmware.
- Std_ReturnType [Eth_43_PFE_GetGpiStats](#) (uint8 u8GpiIndex, Eth_43_PFE_GpiStatsType *stat)
Get gpi statistics from firmware.
- Std_ReturnType [Eth_43_PFE_GetBmuStats](#) (uint8 u8BmuIndex, Eth_43_PFE_BmuStatsType *stat)
Get bmu statistics from firmware.
- Std_ReturnType [Eth_43_PFE_GetCounterValues](#) (uint8 u8CtrlIdx, Eth_CounterType *CounterPtr)
Reads a list with drop counter values of the corresponding controller.
- Std_ReturnType [Eth_43_PFE_GetRxStats](#) (uint8 u8CtrlIdx, Eth_RxStatsType *RxStatsPtr)
Read the status of a controller.
- Std_ReturnType [Eth_43_PFE_GetTxStats](#) (uint8 u8CtrlIdx, Eth_TxStatsType *TxStatsPtr)
Return the list of Transmission Statistics.
- Std_ReturnType [Eth_43_PFE_GetTxErrorCounterValues](#) (uint8 u8CtrlIdx, Eth_TxErrorCounterValues↵Type *TxErrorCounterValuesPtr)
Return the list of Transmission Statistics.
- Std_ReturnType [Eth_43_PFE_ChannelBdFlushRx](#) (pfe_ct_phy_if_id_t slaveHifPhyIfID)
Disables a HIF channel and flushes its cached RX BDs.
- void [Eth_43_PFE_GetVersionInfo](#) (Std_VersionInfoType *VersionInfoPtr)
Returns the version information of this module.
- void [Eth_43_PFE_MainFunction](#) (void)
The function checks for controller errors and lost frames. Used for polling state changes. Calls EthIf_CtrlMode↵Indication when the controller mode changed.
- void * [Eth_43_PFE_GetPlatform](#) (void)
The function can be used to export the pfe_platform instance, which might be needed by external (slave) HIF driver.
- Std_ReturnType [Eth_43_PFE_SetGlobalTime](#) (uint8 u8CtrlIdx, const Eth_TimeStampType *timeStamp↵Ptr)
Set the global time for the EMAC corresponding to provided controller index.
- Std_ReturnType [Eth_43_PFE_SetCorrectionTime](#) (uint8 u8CtrlIdx, const Eth_TimeIntDiffType *time↵OffsetPtr, const Eth_RateRatioType *rateRatioPtr)
Allows the Time Slave to adjust the local ETH Reference clock in HW.
- Std_ReturnType [Eth_43_PFE_GetCurrentTime](#) (uint8 u8CtrlIdx, Eth_TimeStampQualType *timeQual↵Ptr, Eth_TimeStampType *timeStampPtr)
Get the global time from the EMAC corresponding to provided controller index.
- Std_ReturnType [Eth_43_PFE_GetEgressTimeStamp](#) (uint8 u8CtrlIdx, Eth_BufIdxType BufIdx, Eth_↵TimeStampQualType *timeQualPtr, Eth_TimeStampType *timeStampPtr)
Get the egress time stamp of the last packet sending from certain controller.
- Std_ReturnType [Eth_43_PFE_GetIngressTimeStamp](#) (uint8 u8CtrlIdx, const Eth_DataType *DataPtr, Eth_TimeStampQualType *timeQualPtr, Eth_TimeStampType *timeStampPtr)
Get the ingress time stamp of the received packet.
- void [Eth_43_PFE_EnableEgressTimeStamp](#) (uint8 u8CtrlIdx, Eth_BufIdxType BufIdx)
Activate egress time stamping on a dedicated message object.
- void [Eth_43_PFE_SetPhysAddr](#) (uint8 u8CtrlIdx, const uint8 *PhysAddrPtr)
Set or change physical address to the defined controller.
- Std_ReturnType [Eth_43_PFE_GetTmuStats](#) (Eth_43_PFE_TmuStatsType *stat)
Get tmu statistics from firmware.

6.1.2 Function Reference

6.1.2.1 Eth_43_PFE_HifIrqHdlr_0()

```
Eth_43_PFE_HifIrqHdlr_0 ( )
```

Interrupt handler for HIF0 channel IRQ.

This interrupt is shared between all Ethernet interfaces that are connected through HIF0. It handles both received frames and Tx confirmations.

Note

The interrupt is shared among all configured controllers and when it is triggered, it will handle received frames and transmit confirmations for all controllers.

6.1.2.2 Eth_43_PFE_HifIrqHdlr_1()

```
Eth_43_PFE_HifIrqHdlr_1 ( )
```

Interrupt handler for HIF1 channel IRQ.

This interrupt is shared between all Ethernet interfaces that are connected through HIF1. It handles both received frames and Tx confirmations.

Note

The interrupt is shared among all configured controllers and when it is triggered, it will handle received frames and transmit confirmations for all controllers.

6.1.2.3 Eth_43_PFE_HifIrqHdlr_2()

```
Eth_43_PFE_HifIrqHdlr_2 ( )
```

Interrupt handler for HIF2 channel IRQ.

This interrupt is shared between all Ethernet interfaces that are connected through HIF2. It handles both received frames and Tx confirmations.

Note

The interrupt is shared among all configured controllers and when it is triggered, it will handle received frames and transmit confirmations for all controllers.

6.1.2.4 Eth_43_PFE_HifIrqHdlr_3()

Eth_43_PFE_HifIrqHdlr_3 ()

Interrupt handler for HIF3 channel IRQ.

This interrupt is shared between all Ethernet interfaces that are connected through HIF3. It handles both received frames and Tx confirmations.

Note

The interrupt is shared among all configured controllers and when it is triggered, it will handle received frames and transmit confirmations for all controllers.

6.1.2.5 Eth_43_PFE_HifNoCpyIrqHdlr()

Eth_43_PFE_HifNoCpyIrqHdlr ()

Interrupt handler for HIF Nocopy channel IRQ.

This interrupt is shared between all Ethernet interfaces that are connected through HIF Nocopy. It handles both received frames and Tx confirmations.

Note

The interrupt is shared among all configured controllers and when it is triggered, it will handle received frames and transmit confirmations for all controllers.

6.1.2.6 Eth_43_PFE_BmuIrqHdlr()

Eth_43_PFE_BmuIrqHdlr ()

Interrupt handler for BMUs.

This interrupt is shared between both BMUs.

Note

This interrupt is supported only if option Enable BMU Interrupt is enabled.

6.1.2.7 Eth_43_PFE_Transmit()

```
Std_ReturnType Eth_43_PFE_Transmit (
    uint8 u8CtrlIdx,
    Eth_BufIdxType BufIdx,
    Eth_FrameType FrameType,
    boolean TxConfirmation,
    uint16 LenByte,
    const uint8 * PhysAddrPtr )
```

Triggers transmission of a previously filled transmit buffer.

Triggers transmission of a previously granted and then filled transmit buffer. Reentrant for different buffer indexes and Ctrl indexes.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which buffer shall be transmitted. The index is valid within the context of the Ethernet Driver only.
in	<i>BufIdx</i>	Index of the buffer resource to be transmitted.
in	<i>FrameType</i>	Desired value of the Ethernet frame type in the frame header.
in	<i>TxConfirmation</i>	Activates transmission confirmation.
in	<i>LenByte</i>	Buffer data length in bytes (payload length).
in	<i>PhysAddrPtr</i>	Physical target address (MAC address) in network byte order.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed.

6.1.2.8 Eth_43_PFE_ProvideTxBuffer()

```
BufReq_ReturnType Eth_43_PFE_ProvideTxBuffer (
    uint8 u8CtrlIdx,
    uint8 Priority,
    Eth_BufIdxType * BufIdxPtr,
    uint8 ** BufPtr,
    uint16 * LenBytePtr )
```

Provides access to a transmit buffer of the specified controller.

Provides access to a transmit buffer of the FIFO related to the specified priority.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which buffer shall be provided. The index is valid within the context of the Ethernet Driver only.
in	<i>Priority</i>	Frame priority for transmit buffer FIFO selection.
out	<i>BufIdxPtr</i>	Index to the granted transmit buffer resource. It uniquely identifies the buffer in all subsequent calls of functions Eth_43_PFE_Transmit() and Eth_43_PFE_TxConfirmation() .
out	<i>BufPtr</i>	Pointer to the granted buffer. This is the space where the data to be transmitted shall be stored.
in, out	<i>LenBytePtr</i>	Buffer payload length \ • In: desired length in bytes \ • Out: granted length in bytes \

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Returns

Error and buffer status

Return values

<i>BUFREQ_OK</i>	Buffer was successfully granted and no error has occurred.
<i>BUFREQ_E_NOT_OK</i>	A development error was detected and no buffer was granted.
<i>BUFREQ_E_BUSY</i>	All available buffers in use therefore no buffer was granted. No error has been detected.

Note

The application should handle possible difference between the requested and granted buffer lengths. It is not necessary to use whole granted buffer i.e. some space at the end may not be written.

6.1.2.9 Eth_43_PFE_ReleaseTxBuffer()

```
BufReq_ReturnType Eth_43_PFE_ReleaseTxBuffer (
    uint8 u8CtrlIdx,
    Eth_BufIdxType BufIdx )
```

Release a previously granted TX buffer. The transmission must not have been triggered.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller whose buffer shall be released The index is valid within the context of the Ethernet Driver only.
in	<i>BufIdx</i>	Index of the buffer resource to be released.

Returns

Error status

Return values

<i>BUFREQ_E_OK</i>	- Buffer successfully released
<i>BUFREQ_E_NOT_OK</i>	- Development error was detected, no buffer was released.
<i>BUFREQ_E_BUSY</i>	- Couldn't free the TX buffer of an un-finished transmission.

6.1.2.10 Eth_43_PFE_TxConfirmation()

```
void Eth_43_PFE_TxConfirmation (
    uint8 u8CtrlIdx )
```

Triggers frame transmission confirmations.

All transmit buffers are checked and upper layers are informed about successfully transmitted frames. Buffers containing transmitted frames are unlocked after the confirmation.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which shall be checked whether any frame transmission has finished. The index is valid within the context of the Ethernet Driver only.
----	------------------	--

Note

This function can be called only if Tx interrupts are disabled

6.1.2.11 Eth_43_PFE_PreInit()

```
void Eth_43_PFE_PreInit (
    const Eth_43_PFE_ConfigType * CfgPtr )
```

The function should be used to initialize pfe interfaces.

The function must be called before the PFE partition transits from reset (partition 2, configured in Mcu plugin).

Parameters

in	<i>CfgPtr</i>	Points to the implementation specific structure containing the Eth driver configuration. The same configuration pointer shall be used for both Eth_43_PFE_PreInit and Eth_43_PFE_Init
----	---------------	---

6.1.2.12 Eth_43_PFE_Init()

```
void Eth_43_PFE_Init (
    const Eth_43_PFE_ConfigType * CfgPtr )
```

Initializes the Ethernet Driver.

Passed configuration pointer is internally stored and the driver is initialized. The Ethernet controller is not touched.

Note

Function should be called only once.

Warning

Second call can cause undefined behavior. Call the Eth_SetControllerMode() and pass ETH_MODE_DOWN to the CtrlMode argument before the second Eth_43_PFE_Init call to avoid problems.

Parameters

in	<i>CfgPtr</i>	Points to the implementation specific structure containing the Eth driver configuration. The same configuration pointer shall be used for both Eth_43_PFE_PreInit and Eth_43_PFE_Init.
----	---------------	--

6.1.2.13 Eth_43_PFE_DeInit()

```
void Eth_43_PFE_DeInit (
    void )
```

Shutdown the driver gracefully.

Note

In master-slave scenarios, the master should not call Eth_43_PFE_DeInit when slaves are still running. The recommended sequence is to call Eth_43_PFE_DeInit on slaves first, then call Eth_43_PFE_DeInit on the master.

6.1.2.14 Eth_43_PFE_SetControllerMode()

```
Std_ReturnType Eth_43_PFE_SetControllerMode (
    uint8 u8CtrlIdx,
    Eth_ModeType CtrlMode )
```

Enables or disables the given controller.

Disabling the controller clears all receive and transmit buffers. The application should ensure that no data is lost.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller to be enabled or disabled. The index is valid within the context of the Ethernet Driver only.
in	<i>CtrlMode</i>	Mode which shall be entered
158		• ETH_MODE_DOWN: disable the controller • ETH_MODE_ACTIVE: enable the controller S32G PFE MCAL 4.4 Driver NXP Semiconductors

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed.

6.1.2.15 Eth_43_PFE_GetControllerMode()

```
Std_ReturnType Eth_43_PFE_GetControllerMode (
    uint8 u8CtrlIdx,
    Eth_ModeType * CtrlModePtr )
```

Obtains the mode of the given controller.

Mode to which the PFE controller transitioned.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
out	<i>CtrlModePtr</i>	Pointer where to store the current controller mode.

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected and the function failed.

6.1.2.16 Eth_43_PFE_GetPhysAddr()

```
void Eth_43_PFE_GetPhysAddr (
    uint8 u8CtrlIdx,
    uint8 * PhysAddrPtr )
```

Obtains the physical source address used by the indexed controller (the node MAC address).

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which MAC address should be read. The index is valid within the context of the Ethernet Driver only.
out	<i>PhysAddrPtr</i>	Pointer where to store physical source address (MAC address). The address in network byte order is stored into 6 bytes at the given memory address.

6.1.2.17 Eth_43_PFE_UpdatePhysAddrFilter()

```
Std_ReturnType Eth_43_PFE_UpdatePhysAddrFilter (
    uint8 u8CtrlIdx,
    const uint8 * PhysAddrPtr,
    Eth_FilterActionType Action )
```

Adds or removes the specified PhysAddrPtr address to or from a multicast address pool in the controller specified by CtrlIdx.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller. The index is valid within the context of the Ethernet Driver only.
in	<i>PhysAddrPtr</i>	Pointer to PHY address which shall be added or removed to or from multicast pool. The address in network byte order stored into 6 bytes of memory.
in	<i>Action</i>	Determine whenever the defined address will be added to the pool ETH_ADD_TO_FILTER or removed from it ETH_REMOVE_FROM_FILTER.

Note

Each EMAC can handle up to 8 MAC addresses with exact match. All addresses, multicast and unicast, configured in all drivers (in master-slave scenarios) are counted in that. When more than 8 addresses are configured for one EMAC, the EMAC hash table will be used for filtering. This may cause the EMAC to accept also frames with MAC addresses that were not configured (when using hash matching algorithm, all frames with same hash group will be accepted). These additional frames (with DMAC address that was not configured) will be received by master driver, unless default configuration is changed through FCI.

6.1.2.18 Eth_43_PFE_Receive()

```
void Eth_43_PFE_Receive (
    uint8 u8CtrlIdx,
    uint8 FifoIdx,
    Eth_RxStatusType * RxStatusPtr )
```

Triggers frames reception notifications.

All receive buffers are checked and the first received frame is passed to the EthIf module. The caller is notified whether any frame was received and whether more frames are available in the receive queue.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which shall be checked whether any new frames were received. The index is valid within the context of the Ethernet Driver only.
in	<i>FifoIdx</i>	Index of Rx Fifo where frames were received.
out	<i>RxStatusPtr</i>	Informes the caller whether a frame was received (ETH_RECEIVED or ETH_NOT_RECEIVED) and whether more frames are available in the queue (ETH_RECEIVED or ETH_RECEIVED_MORE_DATA_AVAILABLE).

6.1.2.19 Eth_43_PFE_WriteMii()

```
Std_ReturnType Eth_43_PFE_WriteMii (
    uint8 u8CtrlIdx,
    uint8 u8TrcvIdx,
    uint8 u8RegIdx,
    uint16 u16RegVal )
```

Write specified transceiver register through the MII.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller withing the context of the Ethernet Driver
in	<i>u8TrcvIdx</i>	Index of the transceiver on the MII
in	<i>u8RegIdx</i>	Index of the transceiver register on the MII
in	<i>u16RegVal</i>	Value to be written into the indexed register

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.20 Eth_43_PFE_ReadMii()

```
Std_ReturnType Eth_43_PFE_ReadMii (
    uint8 u8CtrlIdx,
    uint8 u8TrcvIdx,
    uint8 u8RegIdx,
    uint16 * pu16RegValPtr )
```

Read the specified transceiver register through the MII.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller withing the context of the Ethernet Driver
in	<i>u8TrcvIdx</i>	Index of the transceiver on the MII
in	<i>u8RegIdx</i>	Index of the transceiver register on the MII
out	<i>pu16RegValPtr</i>	Filled with the register content of the indexed register

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.21 Eth_43_PFE_WriteMii45()

```
Std_ReturnType Eth_43_PFE_WriteMii45 (
    uint8 u8CtrlIdx,
    uint8 u8TrcvIdx,
    uint8 u8DevIdx,
    uint16 u16RegIdx,
    uint16 u16RegVal )
```

Write specified transceiver register through the MII (Clause 45)

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller withing the context of the Ethernet Driver
in	<i>u8TrcvIdx</i>	Index of the transceiver
in	<i>u8DevIdx</i>	Index of the device within the transceiver
in	<i>u16RegIdx</i>	Index of the transceiver register
in	<i>u16RegVal</i>	Value to be written into the indexed register

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.22 Eth_43_PFE_ReadMii45()

```
Std_ReturnType Eth_43_PFE_ReadMii45 (
    uint8 u8CtrlIdx,
```

```

uint8 u8TrcvIdx,
uint8 u8DevIdx,
uint16 u16RegIdx,
uint16 * pu16RegValPtr )

```

Read the specified transceiver register through the MII (Clause 45)

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller withing the context of the Ethernet Driver
in	<i>u8TrcvIdx</i>	Index of the transceiver (address)
in	<i>u16RegIdx</i>	Index of the transceiver register
in	<i>u8DevIdx</i>	Index of the device within the transceiver
out	<i>pu16RegValPtr</i>	Filled with the register content of the indexed register

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.23 Eth_43_PFE_GetClassStats()

```

Std_ReturnType Eth_43_PFE_GetClassStats (
    pfe_ct_classify_stats_t * stat )

```

Get class statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
-----	-------------	---------------------

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to allocate memory for read

6.1.2.24 Eth_43_PFE_GetRtableStats()

```

Std_ReturnType Eth_43_PFE_GetRtableStats (
    pfe_ct_conntrack_stats_t * stat,
    uint8 conntrack_index )

```



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Get rtable statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
out	<i>conntrack_index</i>	conntrack index

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.25 Eth_43_PFE_GetL2BridgeDomainStats()

```
Std_ReturnType Eth_43_PFE_GetL2BridgeDomainStats (
    pfe_ct_vlan_stats_t * stat,
    uint32 index_vlan )
```

Get l2 bridge statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
in	<i>index_vlan</i>	index of vlan

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.26 Eth_43_PFE_GetL2BridgeStats()

```
Std_ReturnType Eth_43_PFE_GetL2BridgeStats (
    Eth_43_PFE_L2BridgeStatsType * stat,
    uint32 index_entry )
```

Get l2 bridge statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
-----	-------------	---------------------

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.27 Eth_43_PFE_GetWdtStats()

```
Std_ReturnType Eth_43_PFE_GetWdtStats (
    Eth_43_PFE_WdtStatsType * stat )
```

Get wdt statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
-----	-------------	---------------------

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.28 Eth_43_PFE_GetGpiStats()

```
Std_ReturnType Eth_43_PFE_GetGpiStats (
    uint8 u8GpiIndex,
    Eth_43_PFE_GpiStatsType * stat )
```

Get gpi statistics from firmware.

Parameters

in	<i>u8GpiIndex</i>	Gpi instance
out	<i>stat</i>	Statistic structure

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.29 Eth_43_PFE_GetBmuStats()

```
Std_ReturnType Eth_43_PFE_GetBmuStats (
    uint8 u8BmuIndex,
    Eth_43_PFE_BmuStatsType * stat )
```

Get bmu statistics from firmware.

Parameters

in	<i>u8BmuIndex</i>	Bmu instance
out	<i>stat</i>	Statistic structure

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Not possible to get statistics

6.1.2.30 Eth_43_PFE_GetCounterValues()

```
Std_ReturnType Eth_43_PFE_GetCounterValues (
    uint8 u8CtrlIdx,
    Eth_CounterType * CounterPtr )
```

Reads a list with drop counter values of the corresponding controller.

Reads a list with drop counter values of the corresponding controller. The meaning of these values is hardware dependent. However, the list DropCount[] shall contain the following values in the given order, where the maximal possible value shall denote an invalid value, e.g. if this counter is not available:

1. Dropped packets due to buffer overrun
2. Dropped packets due to CRC errors
3. Number of undersize packets which were less than 64 octets long (excluding framing bits, but including FCS octets) and were otherwise well formed. (see IETF RFC 1757)
4. Number of oversize packets which are longer than 1518 octets (excluding framing bits, but including FCS octets) and were otherwise well formed. (see IETF RFC 1757)
5. Number of alignment errors, i.e. packets which are received and are not an integral number of octets in length and do not pass the CRC.
6. SQE test error according to IETF RFC1643 dot3StatsSQETestErrors
7. The number of inbound packets which were chosen to be discarded even though no errors had been detected to prevent their being deliverable to a higher layer protocol. One possible reason for discarding such a packet could be to free up buffer space. (see IETF RFC 2233 ifInDiscards)

8. Total number of erroneous inbound packets
9. The number of outbound packets which were chosen to be discarded even though no errors had been detected to prevent their being transmitted. One possible reason for discarding such a packet could be to free up buffer space. (see IETF RFC 2233 ifOutDiscards)
10. Total number of erroneous outbound packets
11. Single collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by exactly one collision. (see IETF RFC1643 dot3StatsSingleCollisionFrames)
12. Multiple collision frames: A count of successfully transmitted frames on a particular interface for which transmission is inhibited by more than one collision. (see IETF RFC1643 dot3StatsMultipleCollisionFrames)
13. Number of deferred transmission: A count of frames for which the first transmission attempt on a particular interface is delayed because the medium is busy. (see IETF RFC1643 dot3StatsDeferredTransmissions)
14. Number of late collisions: The number of times that a collision is detected on a particular interface later than 512 bit times into the transmission of a packet. (see IETF RFC1643 dot3StatsLateCollisions)
15. The following positions in the list can contain hardware dependent counter values

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller
out	<i>CounterPtr</i>	Pointer to Eth_CounterType where to store the read drop counter values

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected or inaccessible to counters register and the function.

6.1.2.31 Eth_43_PFE_GetRxStats()

```
Std_ReturnType Eth_43_PFE_GetRxStats (
    uint8 u8CtrlIdx,
    Eth_RxStatsType * RxStatsPtr )
```

Read the status of a controller.

Returns the following list according to IETF RFC2819, where the maximal possible value shall denote an invalid value, e.g. if this counter is not available:

1. etherStatsDropEvents

2. etherStatsOctets
3. etherStatsPkts
4. etherStatsBroadcastPkts
5. etherStatsMulticastPkts
6. etherStatsCrcAlignErrors
7. etherStatsUndersizePkts
8. etherStatsOversizePkts
9. etherStatsFragments
10. etherStatsJabbers
11. etherStatsCollisions
12. etherStatsPkts64Octets
13. etherStatsPkts65to127Octets
14. etherStatsPkts128to255Octets
15. etherStatsPkts256to511Octets
16. etherStatsPkts512to1023Octets
17. etherStatsPkts1024to1518Octets

Parameters

in	<i>CtrlIdx</i>	Index of the controller which shall be read the status register.
out	<i>RxStatsPtr</i>	Pointer to 32 bit long memory space to be filled with the list values according to IETF RFC 2819 (Remote Network Monitoring Management Information Base).

Returns

Error status

Return values

<i>E_OK</i>	No error was detected during the function execution.
<i>E_NOT_OK</i>	Development error was detected or inaccessible to counters register and the function.

6.1.2.32 Eth_43_PFE_GetTxStats()

```
Std_ReturnType Eth_43_PFE_GetTxStats (
    uint8 u8CtrlIdx,
    Eth_TxStatsType * TxStatsPtr )
```

Return the list of Transmission Statistics.

Parameters

in	<i>CtrlIdx</i>	Index of controller within the context of the Ethernet Driver.
out	<i>TxStatsPtr</i>	Pointer to 32 bit long memory space to be filled with the list values according to IETF RFC 1213

Return the list of Transmission Statistics out of IETF RFC 1213 defined within Eth_TxStatsType, where the maximal possible value shall denote an invalid value. e.g. this counter is not available..

Returns

The search status

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Tx-statistics could not be obtained.

6.1.2.33 Eth_43_PFE_GetTxErrorCounterValues()

```
Std_ReturnType Eth_43_PFE_GetTxErrorCounterValues (
    uint8 u8CtrlIdx,
    Eth_TxErrorCounterValuesType * TxErrorCounterValuesPtr )
```

Return the list of Transmission Statistics.

Return the list of Transmission Error Counters out of IETF RFC1213 and RFC1643 defined within Eth_TxErrorCounterValuesType, where the maximal possible value shall denote an invalid value. e.g. this counter is not available.

Parameters

in	<i>CtrlIdx</i>	Index of controller within the context of the Ethernet Driver.
out	<i>TxErrorCounterValuesPtr</i>	List of values to read statistic error counter values for transmission.

Returns

The search status

Return values

<i>E_OK</i>	Success
<i>E_NOT_OK</i>	Tx-statistics could not be obtained.

6.1.2.34 Eth_43_PFE_ChannelBdFlushRx()

```
Std_ReturnType Eth_43_PFE_ChannelBdFlushRx (
    pfe_ct_phy_if_id_t slaveHifPhyIfID )
```

Disables a HIF channel and flushes its cached RX BDs.

The function is intended for assisting a slave driver to recover from crash. It:

- disables the HIF channel, so it will no longer receive normal traffic,
- flushes cached Rx buffer descriptors from the HIF channel. During this process the PFE HW needs write access to buffers and buffer descriptor memory which was previously configured by the crashed driver.

Note

This function can only be used for HIF channel of a slave driver which is no longer running.

This function is available only on master driver, only if option "Enable PFE Channel BD Flush API" is enabled.

Parameters

in	<i>slaveHifPhyIfID</i>	Physical interface ID of the HIF used by the slave driver which has crashed.
----	------------------------	--

Return values

<i>E_OK</i>	Flushing was successfull and is complete
<i>E_NOT_OK</i>	Flushing has failed
<i>ETH_43_PFE_E_AGAIN</i>	Another call of this function is needed to continue the flush (due to limit configured in option <i>PfeChannelBdFlushMaxBdCount</i>)

6.1.2.35 Eth_43_PFE_GetVersionInfo()

```
void Eth_43_PFE_GetVersionInfo (
    Std_VersionInfoType * VersionInfoPtr )
```

Returns the version information of this module.

Parameters

out	<i>VersionInfoPtr</i>	Pointer where to store the version information of this particular module instance.
-----	-----------------------	--

6.1.2.36 Eth_43_PFE_MainFunction()

```
void Eth_43_PFE_MainFunction (
    void )
```

The function checks for controller errors and lost frames. Used for polling state changes. Calls EthIf_CtrlModeIndication when the controller mode changed.

6.1.2.37 Eth_43_PFE_GetPlatform()

```
void* Eth_43_PFE_GetPlatform (
    void )
```

The function can be used to export the pfe_platform instance, which might be needed by external (slave) HIF driver.

6.1.2.38 Eth_43_PFE_SetGlobalTime()

```
Std_ReturnType Eth_43_PFE_SetGlobalTime (
    uint8 u8CtrlIdx,
    const Eth_TimeStampType * timeStampPtr )
```

Set the global time for the EMAC corresponding to provided controller index.

If option IEEE1588ClockAttachedToGMAC or IEEE1588ClockAttachedToEMAC0 is enabled for underlying EMAC, this function will fail.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller within the context of the Ethernet Driver
in	<i>timeStampPtr</i>	Struct of time value to be configured

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.39 Eth_43_PFE_SetCorrectionTime()

```
Std_ReturnType Eth_43_PFE_SetCorrectionTime (
    uint8 u8CtrlIdx,
    const Eth_TimeIntDiffType * timeOffsetPtr,
    const Eth_RateRatioType * rateRatioPtr )
```

Allows the Time Slave to adjust the local ETH Reference clock in HW.

Only use this function when this controller is used as Time Slave. If option IEEE1588ClockAttachedToGMAC or IEEE1588ClockAttachedToEMAC0 is enabled for underlying EMAC, this function will fail.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which time shall be corrected
in	<i>timeOffsetPtr</i>	Offset between time stamp grandmaster and time stamp by local clock.
in	<i>rateRatioPtr</i>	Time elements to calculate and to modify the ratio of the frequency of the grandmaster in relation to the frequency of the Local Clock.

6.1.2.40 Eth_43_PFE_GetCurrentTime()

```
Std_ReturnType Eth_43_PFE_GetCurrentTime (
    uint8 u8CtrlIdx,
    Eth_TimeStampQualType * timeQualPtr,
    Eth_TimeStampType * timeStampPtr )
```

Get the global time from the EMAC corresponding to provided controller index.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller within the context of the Ethernet Driver
out	<i>timeQualPtr</i>	Pointer to quality of HW time stamp
out	<i>timeStampPtr</i>	Struct of time value to be read

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.41 Eth_43_PFE_GetEgressTimeStamp()

```
Std_ReturnType Eth_43_PFE_GetEgressTimeStamp (
    uint8 u8CtrlIdx,
```

```
Eth_BufIdxType BufIdx,
Eth_TimeStampQualType * timeQualPtr,
Eth_TimeStampType * timeStampPtr )
```

Get the egress time stamp of the last packet sending from certain controller.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller within the context of the Ethernet Driver
in	<i>BufIdx</i>	Index of the buffer resource was used to be transmitted
out	<i>timeQualPtr</i>	Pointer to quality of HW time stamp
out	<i>timeStampPtr</i>	Struct of time value to be read

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.42 Eth_43_PFE_GetIngressTimeStamp()

```
Std_ReturnType Eth_43_PFE_GetIngressTimeStamp (
    uint8 u8CtrlIdx,
    const Eth_DataType * DataPtr,
    Eth_TimeStampQualType * timeQualPtr,
    Eth_TimeStampType * timeStampPtr )
```

Get the ingress time stamp of the received packet.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller within the context of the Ethernet Driver
in	<i>DataPtr</i>	Pointer to the received packet
out	<i>timeQualPtr</i>	Pointer to quality of HW time stamp
out	<i>timeStampPtr</i>	Struct of time value to be read

Return values

<i>E_OK</i>	Service accepted
<i>E_NOT_OK</i>	Service denied

6.1.2.43 Eth_43_PFE_EnableEgressTimeStamp()

```
void Eth_43_PFE_EnableEgressTimeStamp (
    uint8 u8CtrlIdx,
    Eth_BufIdxType BufIdx )
```

Activate egress time stamping on a dedicated message object.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller within the context of the Ethernet Driver
in	<i>BufIdx</i>	Index of the buffer resource to be used

6.1.2.44 Eth_43_PFE_SetPhysAddr()

```
void Eth_43_PFE_SetPhysAddr (
    uint8 u8CtrlIdx,
    const uint8 * PhysAddrPtr )
```

Set or change physical address to the defined controller.

Parameters

in	<i>u8CtrlIdx</i>	Index of the controller which PHY address should be changed. The index is valid within the context of the Ethernet Driver only.
in	<i>PhysAddrPtr</i>	Pointer to PHY address which should be set to the controller. The address is stored in 6 bytes of memory in network byte order. @caution This function may be called only when the controller is down. Call of function Eth_43_PFE_ControllerInit change MAC address to the default value!

6.1.2.45 Eth_43_PFE_GetTmuStats()

```
Std_ReturnType Eth_43_PFE_GetTmuStats (
    Eth_43_PFE_TmuStatsType * stat )
```

Get tmu statistics from firmware.

Parameters

out	<i>stat</i>	Statistic structure
-----	-------------	---------------------

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Return values

<i>EOK</i>	If possible to get statistics, otherwise E_NOT_OK
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